

Transformable Footwear

Pedal-Ready

A removeable attachment for office shoes to enable wearer the use of cycling cleats.



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Initial Exploration

Transformable **footwear**
for multiple applications

This initial exploration is based on the domain which our group selected initially and multiple concepts were then generated from this.

Why this?

Space Saving

For those who travel or commute often, bringing multiple pairs of shoes can be impractical. A single pair that can adapt to different uses saves space in luggage, backpacks, or under-the-desk storage.

Sustainability

By encouraging the idea of one adaptable product, we would promote a more sustainable approach to fashion. Transformable footwear supports a shift from quick and overconsumption to long-lasting, versatile choices.

Futuristic Appeal

Today's consumers are drawn to innovation that offers both function and style. Transformable footwear taps into that interest, blending utility with an appeal that feels forward-looking and in step with modern lifestyles.

Reducing Overconsumption

Transformable footwear could aim to reduce the demand for multiple pairs, cutting down waste by offering a multi-functional alternative. With fewer shoes required, there's less environmental impact.

23 Billion shoes are
bought annually ^[1]

22 Billion end up as
landfill waste ^[1]

Brainstorming

Sports

Weather Shield
Climbing to Walking
Interchangeable Soles
Cycling to Office
Breathable To
Waterproof
Compression
Adjustments

Aesthetics

3D Printable
Customisation
LED Components
Interchangeable Panels

Comfort

Adjustable Cushioning
Transformable Arch
Support
Temperature Regulation
Dynamic Flexibility

Convenience

Climate Adaptive
Packable / Foldable for
Travel
Detachable Heels
Indoor to Outdoor

Manual

Sliding Mechanism
Folding
Magnetic Panels
Velcro
Zippers
Snap Buttons

Automatic

Heat Reactive
Hydro Responsive
Pressure Sensors
UV Reactive Materials
Electroactive Polymers
Light-Sensitive Materials
Motion Sensitive
Electromagnets

I decided to focus on commuters

Because they're the ones who really need to save space and make their footwear work in two different worlds—outdoors and the office. The largest group of non-vehicle commuters are cyclists, followed closely by people who walk. Both need sturdy, weather-ready shoes to get around, but those same shoes often feel out of place once they're sitting

Biking shoes, for example, are great for grip and comfort on the pedals but look pretty clunky in a professional setting and walkers face a similar issue. Shoes that keep them comfortable on city streets aren't always suited for an office at a desk or heading into a meeting.

Group Mindmap

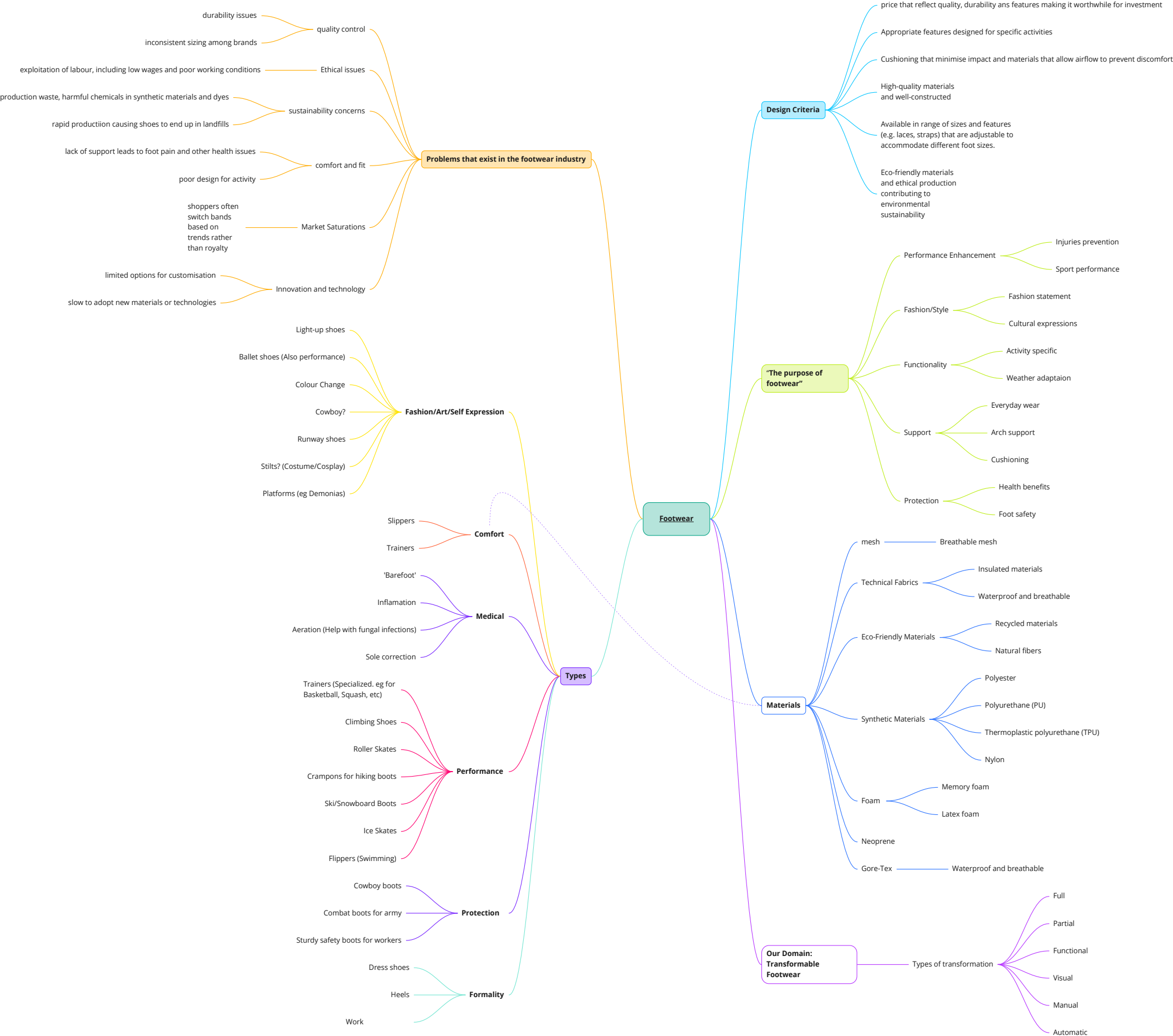
Ruaridh, Jenny, New, Kael, Yousuf

Our group mind map was made using Miro

This mapping helped us explore common problems like discomfort, limited durability, and lack of inclusivity, alongside the varied purposes of footwear—from protection and comfort to style, athletic needs, and therapeutic support.

We also categorized different types of footwear, from athletic and formal shoes to sandals and medical footwear, examining each type’s unique features. This analysis guided us to consider who would use each type of footwear and why, allowing each member to brainstorm ideas tailored to specific user needs.

This helped us to explore our individual ideas further using 5Ws & 1H.



Analysis

Insight

Commuters lead busy lives and often prefer to travel light. They face the challenge of travelling efficiently while maintaining a professional appearance. Since they frequently use multiple forms of transport, such as walking, cycling, and public transit, they require footwear that can accommodate these varying demands.

How might we?

How might we design **footwear** that can transition from a durable weather resistant walking shoe to a professional office shoe?

How might we create **footwear** that can work for cyclists during their commute but transform for office hours?

How might we create workplace **footwear** that can be packed small and light for commuters to carry?



[30]



[32]



[31]

When?

Primarily during morning and evening commutes, when workers are transitioning between different environments—streets, bikes, and offices.

Where?

Urban areas with high commuter populations, where individuals need to balance practicality and style, especially in environments with unpredictable weather or long distances.

Why?

Commuters often face the inconvenience of carrying extra pairs of shoes or wearing uncomfortable footwear for part of their journey. Transformable shoes would offer a practical solution, saving time and effort.

What?

Transformable footwear that serves as a functional commuting shoe (e.g., waterproof for walking, cycling shoes) and can adapt to the professional look required at the workplace.

Who?

Office workers who commute, particularly those who walk, bike, or use public transportation to get to work.

How?

By creating footwear with flexible components (e.g., removable layers, foldable designs, or adjustable parts) that allow easy transformation from a functional commuting shoe to a professional-looking shoe suitable for the office.

Sketch Exploration

Sketching Practice see Page 18

Masculine Office Shoe [5] [6]

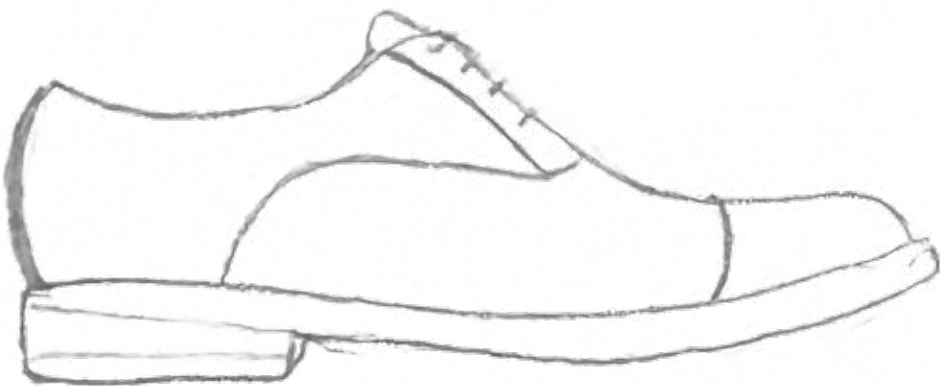
A masculine office shoe is designed to combine professionalism with comfort, typically in a classic Oxford style. Its purpose is to maintain a polished, structured look that pairs well with formalwear.

Structured Upper
Fairly rigid, to hold shape and adds a sharp, refined appearance.

Outsole
Durable, often leather or rubber for more grip and water resistance.

Heel Counter
Reinforced to add stability and helps keep the shoe's shape.

Minimal Details
An undecorated shoe being more formal than a shoe with ornamentation



Feminine Office Shoe [4]

A feminine office shoe balances elegance with comfort, often in styles like pumps or loafers. The design prioritizes a sleek appearance suitable for formal settings.

Structured Upper
Slim and fitted to maintain a refined shape, complementing formal attire while holding structure.

Outsole
Durable, typically leather for a classic appearance, or rubber for extra grip and water resistance on smoother surfaces.

Heel Counter
Reinforced particularly in heeled options, while helping maintain the shoe's shape over time.

Minimal Details
Clean, understated designs, often with a plain or low-profile toe, are considered more formal and versatile for the workplace.



Trainer [2]

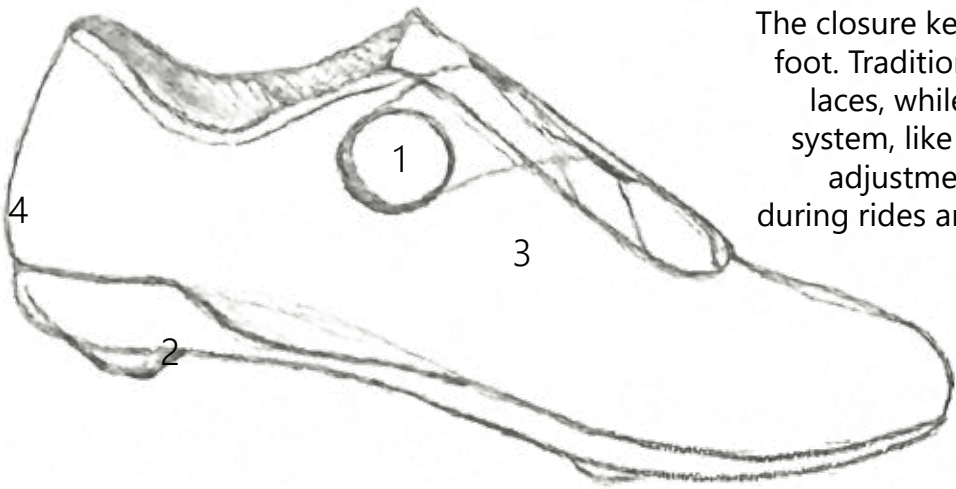
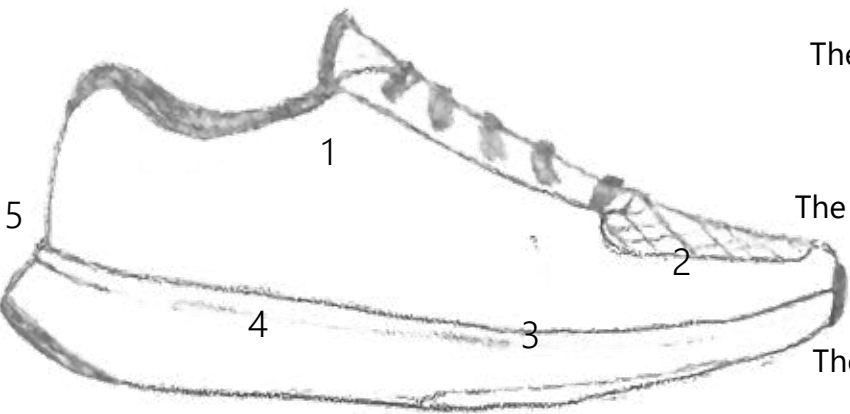
1. Upper
The upper keeps the foot in place, should be breathable though durable.

2. Toe-box
A roomy toe box allows toes to splay, reducing the risk of blisters.

3. Outsole
The outsole is the bottom of the shoe, made of rubber for grip and traction.

4. Midsole
The midsole provides cushioning, often made of foam; thicker midsoles offer more padding.

5. Heel Counter
The heel counter stabilizes the heel and should be firm to handle impact forces.



1. Closure
The closure keeps the shoe snug around the foot. Traditional options include Velcro and laces, while many newer shoes use a dial system, like the BOA® System, for precise adjustments. Closures should stay tight during rides and avoid getting caught in the bike's drivetrain.

2. Cleats
Cleats connect the shoe to the pedal, helping transfer power more efficiently. Road cyclists commonly use three-hole (Look) cleats, which stabilize the foot and improve energy transfer with each pedal stroke.

3. Upper
The upper is the outer layer of the shoe, made from synthetic leather with rubber reinforcements for durability. It should resist water while allowing airflow to keep feet comfortable.

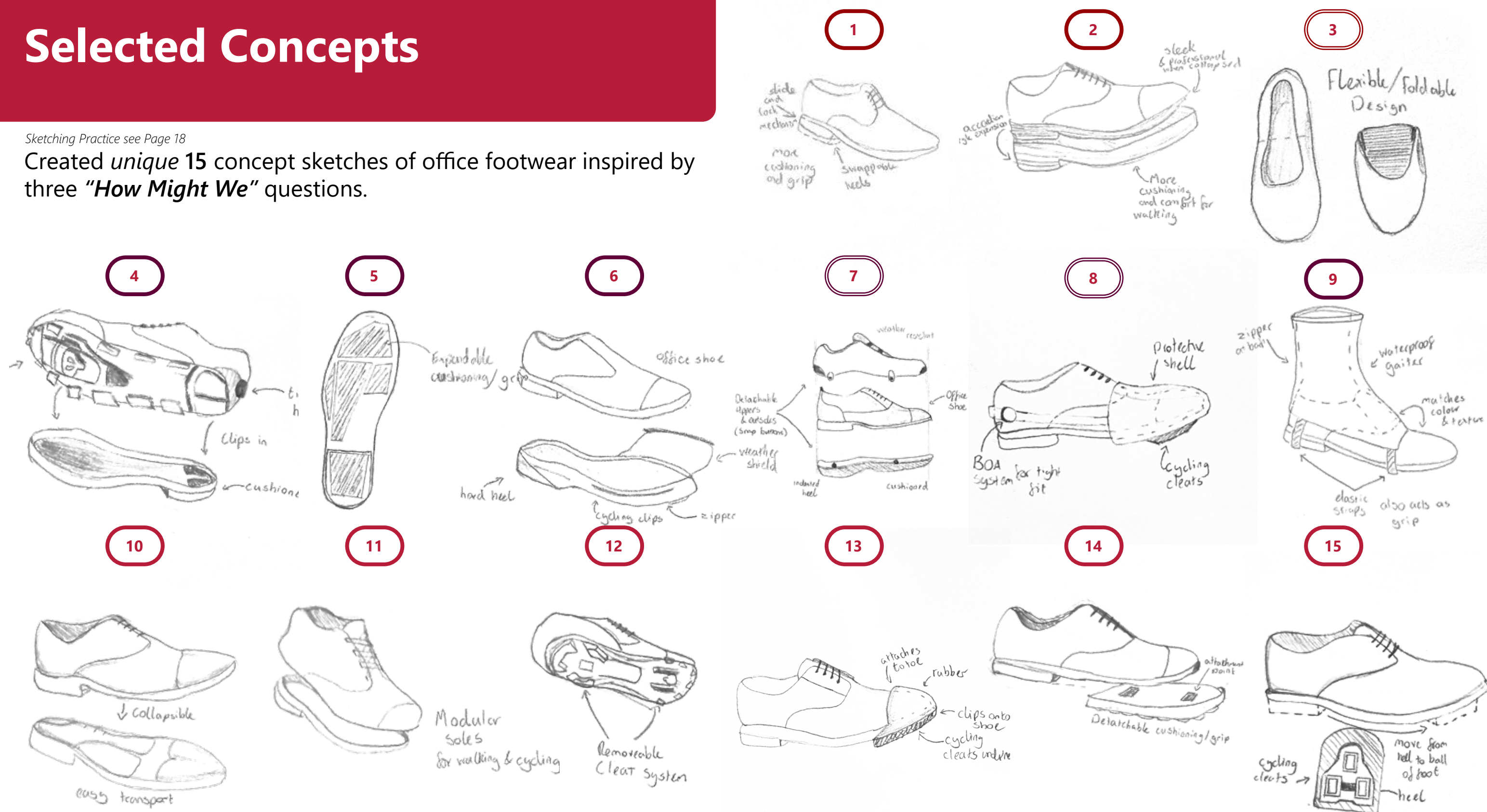
4. Heel Cup
The heel cup holds the heel securely, fitting around the Achilles tendon to prevent the foot from lifting during pedaling. This helps with better energy transfer and a more stable ride.

Road Cycling [3]

Selected Concepts

Sketching Practice see Page 18

Created *unique* 15 concept sketches of office footwear inspired by three “*How Might We*” questions.



How might we design **footwear** that can transition from a durable weather resistant walking shoe to a professional office shoe?

How might we create **footwear** that can work for cyclists during their commute but transform for office hours?

How might we create workplace **footwear** that can be packed small and light for commuters to carry?

Assessing Concepts

These ideas were assessed and narrowed down to three selected concepts based on the criteria of feasibility, desirability, and usability.

How might we design **footwear** that can transition from a durable weather resistant walking shoe to a professional office shoe

1 2 5 7 9 11 14

How might we create **footwear** that can work for cyclists during their commute but transform for office hours

4 6 8 11 12 13 15

How might we create workplace **footwear** that can be packed small and light for commuters to carry

3 10

Feasibility: This criteria assessed the technical practicality of each design. Factors such as manufacturing processes, material availability, cost-efficiency, and durability were considered to determine whether the concept could be realistically produced.

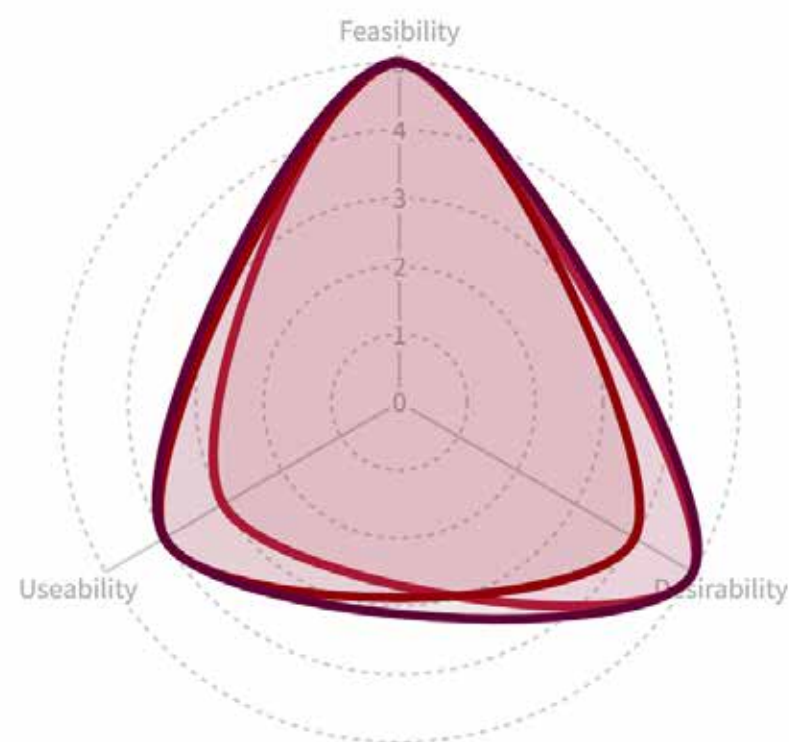
Desirability: This focused on the “*wow factor*” of the design and how useful it would be to potential users. I evaluated each concept based on its ability to impress and its relevance to everyday needs. Additionally, I considered how many people would likely benefit from or be interested in the design.

Usability: This measured the functionality and adaptability of the design in various contexts. The evaluation included ease of use, comfort, and versatility to ensure the concept could cater to a broad range of needs and scenarios.

After scoring each concept against these criteria, the **three** highest-ranking designs were selected.

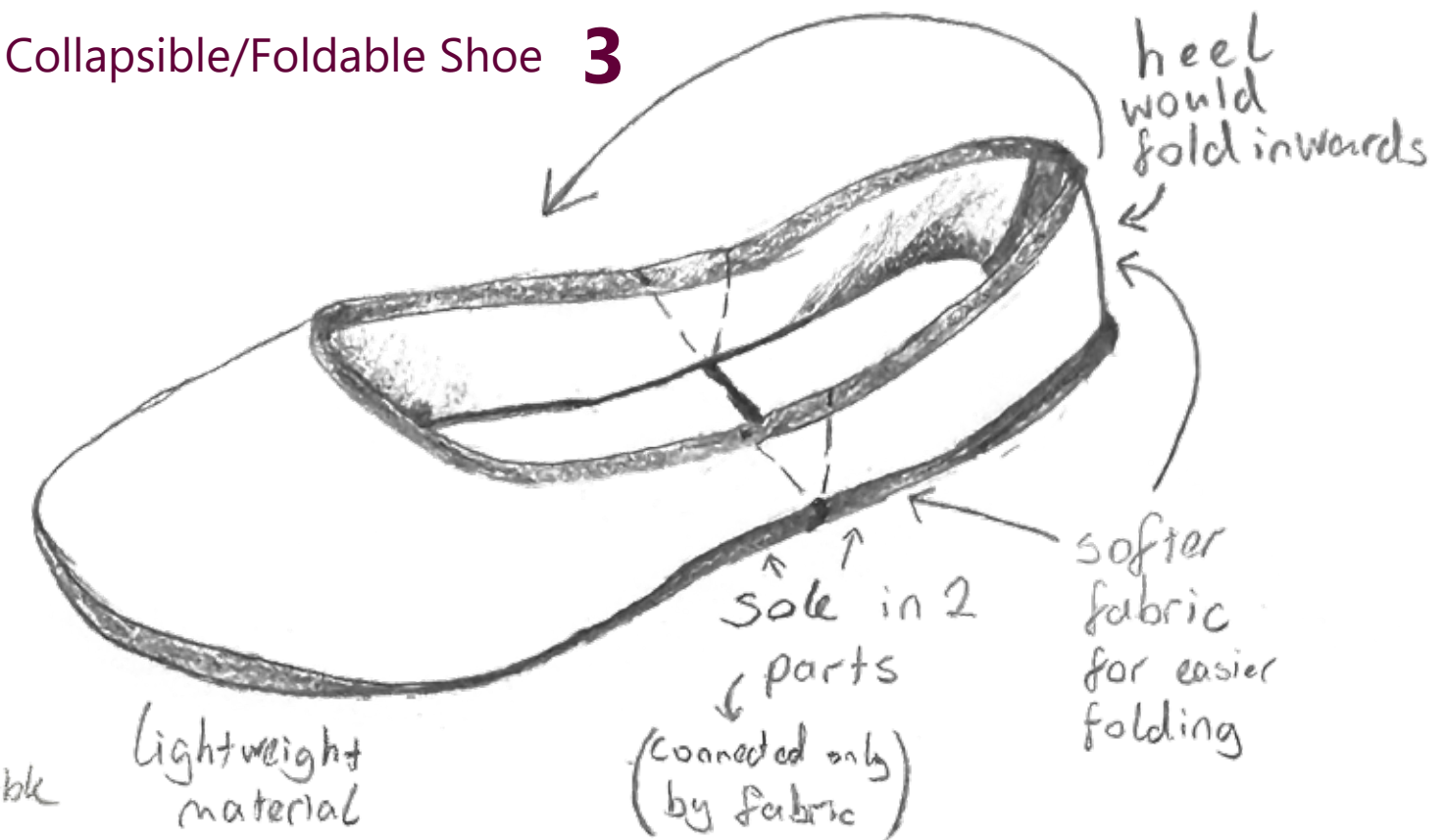
Number	Name	Feasibility	Desirability	Useability	Total
1	Swappable Heels	4	3	4	11
2	Accordion Expansion Sole	2	4	4	10
3	Foldable Shoe	5	5	3	13
4	Cycling Shoe with Cushioned Sole	4	5	3	12
5	Expandable Cushioning	2	3	2	7
6	Office Shoe with Cycling Sole	4	4	3	11
7	Weather Resistant Layer	5	4	4	13
8	Cycling Toe with BOA Strap	5	5	4	14
9	Waterproof Gaiter	5	2	4	11
10	Collapsible Shoe	4	4	4	12
11	Modular Soles	3	4	3	10
12	Removeable Cleats	3	5	3	11
13	Cycling Toe Attachment	3	5	4	12
14	Detachable Cushioning / Grip	4	3	3	10
15	Moveable Heel Cleats	2	5	4	11

■ Foldable Shoe
 ■ Weather Resistant Layer
 ■ Cycling Toe with BOA Strap



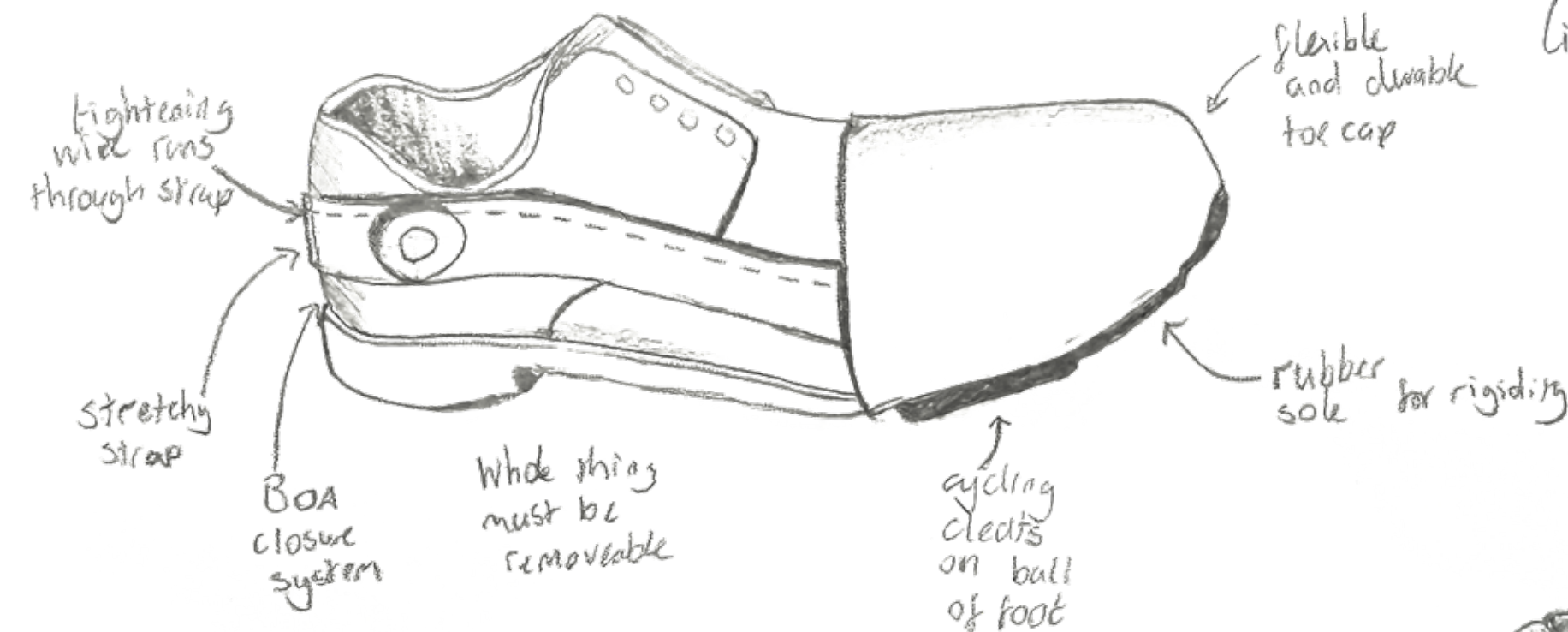
Developing Selected Concepts

Collapsible/Foldable Shoe 3



7 Cycling Toecap With Heel Strap

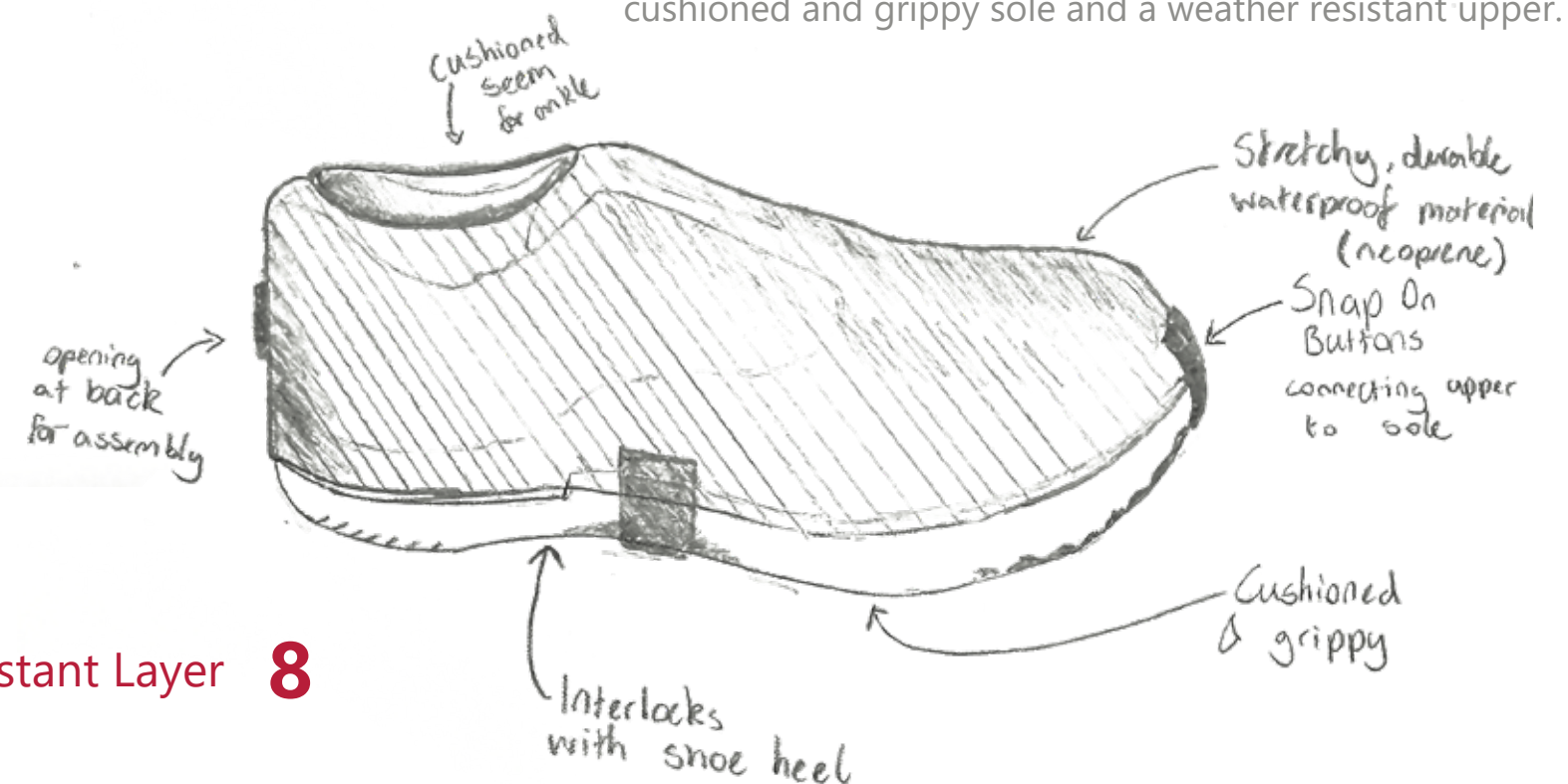
Would be an attachment for any regular office shoe for cycling to and from work.



Would be tightened using the BOA system and would have road bike cycling cleats on the sole.

For more information on the BOA system check **Page 23**

Resistant layer that would surround a regular office shoe, with a cushioned and grippy sole and a weather resistant upper.



Resistant Layer 8

Lo-Fi Prototyping

Lo-Fi Prototyping Practice see Page 21 & 23

Cycling Toecap With Heel Strap 7



Toecap:
Needed to be completely **removable** from the shoe. Being **attachable** and **detachable** easily. Had to provide a position for **cycling cleats** while remaining compact.

Toecap Creation:
Began by molding its shape using layered paper strips over toe. Stuffing the shoe with paper for an accurate fit. For rigidity, a small foam plate at the bottom of the toecap. For the cleats, using an **SPD-SL** cleat as a reference, I cut an outline in foam. Positioning it on top of the foam plate.

Heel Strap:
Half its length into the toecap for security. For tightening, replica of the **BOA closure system**. Would pull wire inside the straps for a snug fit.

Learning:
I realized that the strap directly on the heel could create **discomfort**, as the pressure would land on a small area. So I added a large padded panel at the rear as a **cushion**. The panel also made a good location for the **BOA system**.

The toecap and strap would need to adjust to fit **multiple shoe sizes** and shapes, requiring a material that was both **durable** and **elastic**.



Sole:
Used hot wire to shape foam into the **outline** of an office shoe's sole. Then filing down the foam where the heel is. To curve it, I made cuts in the foam and filled gaps with clay.

Foldable Sole:
Had to ensure the back of the shoe would **fold** within the front but realized it wasn't crucial for the back to fit entirely within the fabric of the upper, as it saved minimal space.



3 Collapsible Shoe



Sole Creation:
A stiff mesh fabric was glued over the middle section of a flip flop sole for **support**. Then layers of paper were added for extra strength. Then the sole was cut, while the mesh was left intact to act as a **hinge**.



Learning:
The back of the upper cannot be rigid or it **won't collapse** with the shoe. The folding would cause stress to the material whenever it is folded, **wearing faster** than a conventional shoe.

8 Resistant Layer

Upper:
Used faux leather from a seat cover as it had the **texture** and **flexibility** I wanted. Though it **lacked weather resistance**. I wrapped the leather tightly over the shoe tracing around the **contours**, and an opening for the leg and at the back of the shoe. I cut along the traced lines and made adjustments for fitting. I used offcuts for more structure in the leg hole.



Snap-on buttons were not available to me, therefore I used tape. Positioned on the sides, front, and back, for **ease of use**.

Learning:
Relying on buttons to secure the sole **won't** provide the best **stability**. The prototype resembles a **winter boot** more than a walking shoe, which was not my initial intention. Though it did suggests new possibilities, such as incorporating grippier soles for better traction on ice and adding insulation for warmth. This design would also only work for one shoe size.

Selecting One Concept

Prototyping

Insights from prototyping showed that the toecap could be designed **to expand and fit a variety of shoe sizes** and shapes. Achieving this meant a **flexible material** capable of stretching for different toe box dimensions would be needed. Additionally, a heel pad was added **to reduce pressure** on the back of the user's foot, which provided an idea for the **BOA closure system**. The BOA system was selected for its consistent tightening mechanism, which is already familiar to road cyclists.

Feasibility

The project is technically feasible because it relies on materials and mechanisms that are **readily available** and widely used in existing products. Waterproof and elastic materials for the toecap are commonly used in outdoor gear, to provide durability and protection. The **BOA closure system** [7] is reliable and is already used in cycling shoes, and cleat attachment designs follow industry standards, making the product compatible with most road bike pedals. Therefore the product can be produced and function as intended without requiring significant technical innovation or untested solutions.

Feedback

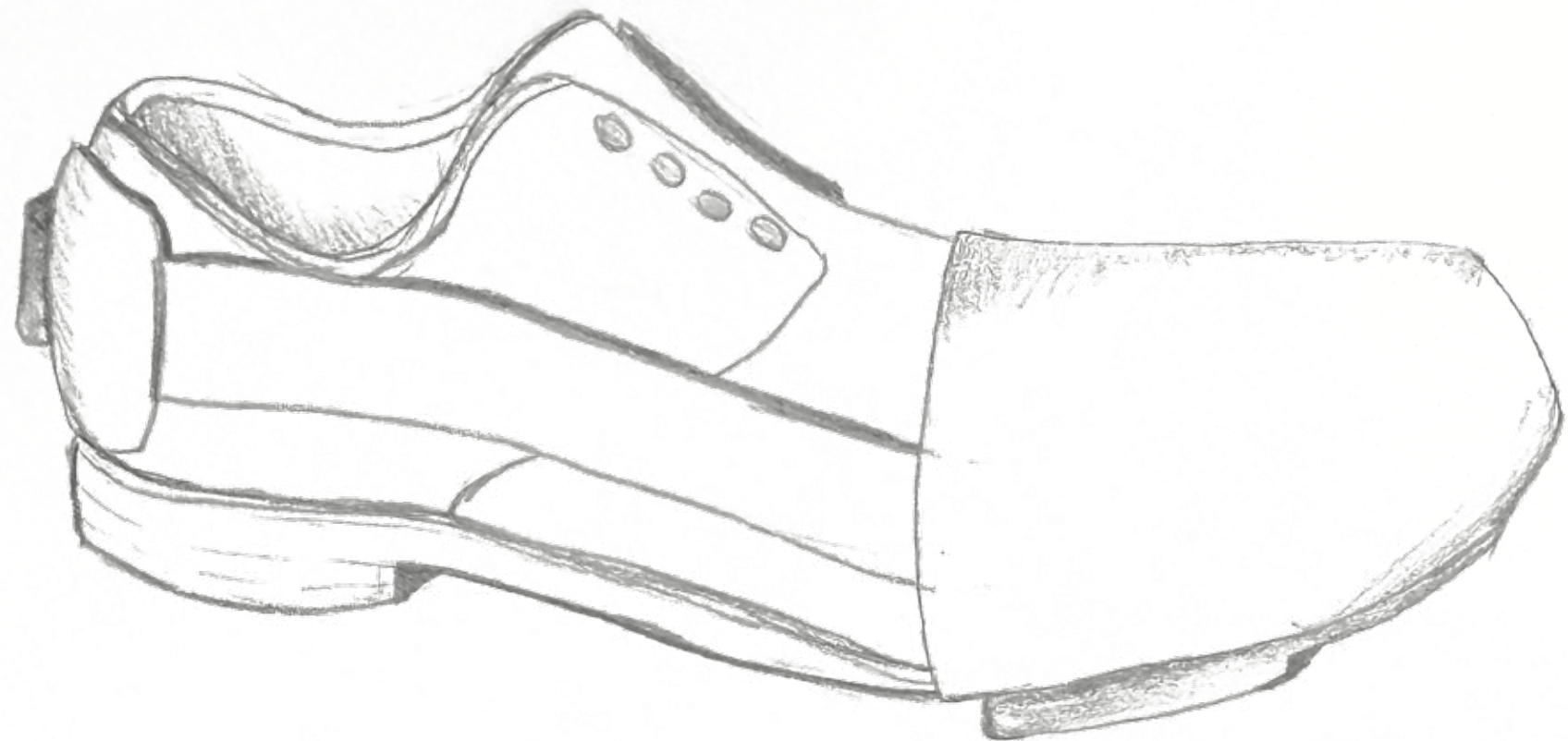
The formative assessment indicated that the concept **addressed the HMW question** of allowing users to cycle comfortably with their everyday work shoes. The feedback supported the overall direction of the project and that the design met the intended goals for **practicality, usability, and adaptability**.

How might we create **footwear** that can work for cyclists during their commute but transform for office hours

Office Shoe Attachment Cycling Toecap With Heel Strap

Featuring a waterproof toecap made from a stretchy neoprene that adapts to different shoe sizes and shapes. The bottom has a rubber for stability and includes integrated cycling cleats for connecting securely to road bikes.

A heel strap wraps around the back of the shoe and is tightened using a BOA closure system, commonly used in cycling footwear, to ensure a firm and reliable fit. This attachment allows users to use their regular shoes for cycling to and from work while being compact and easy to carry.



Requirements & Specifications



SHIMANO Cycling Cleats [14]



Neoprene [19]



BOA Closure System [20]

Requirements

- 1 Sole must connect to bike pedals securely.
- 2 Strap must be able to tighten and loosen in order to remove.
- 3 Must be able to fit different shoe sizes.
- 4 Must be able to be stored compactly.
- 5 Must not damage the original shoe.
- 6 Toebox must be made of stretchy and durable material.
- 7 Must not exert too much pressure on wearer.
- 8 Must not weigh too much.
- 9 Strap must be durable and securely attach.

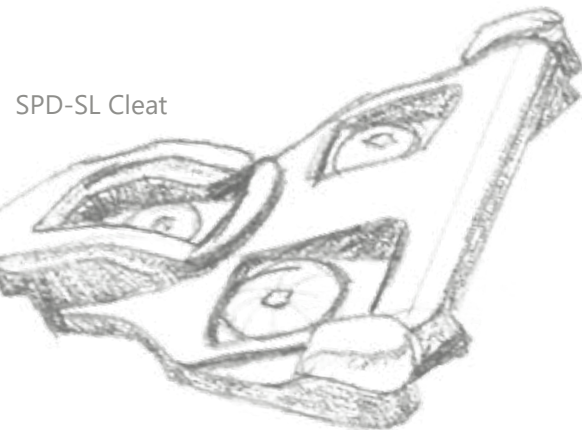
Specification

- Include **cycling cleats** made from a durable material.
- Use a **BOA lacing system** in the straps with micro-adjustments, made up of a steel lace for durability.
- Fit shoe sizes **UK 5-12** with stretchable material and adjustable strap design.
- Strap should fit inside the volume of the toecap. The material should be **flexible** but **maintains shape** after repeated folding.
- No **hard** or **abrasive** materials must come in contact with the original shoe to prevent scratches or marks on the shoe.
- Use **neoprene** [8] or elastic material with abrasion resistance.
- Distribute pressure evenly using **padding** at pressure points such as back of foot.
- Attachment must have a mass of less than **250 grams** per foot; materials should retain very minimal water during rainfall.
- Wires must be made from reinforced **nylon** to withstand force; and reinforced stitching at connection points.

Sketch Analysis

Sketch analysis of the three most important requirements for the project

Sole must connect to bike pedals securely

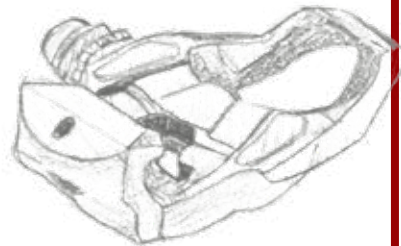


SPD-SL Cleat

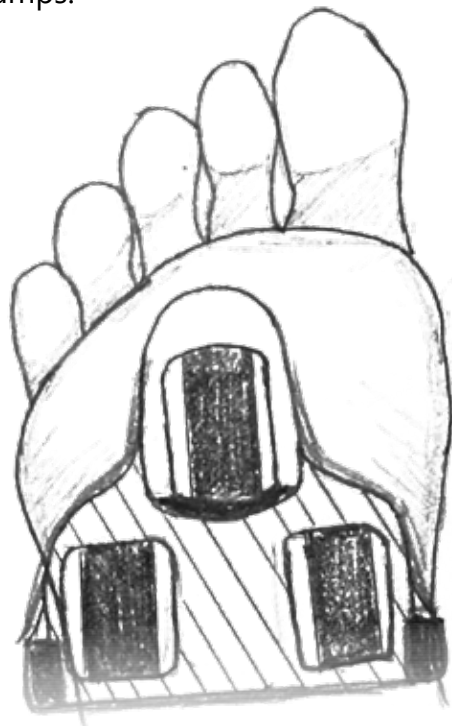
Using **cleats** we can attach the sole to the bike pedal securely. With just flat pedals, the contact with the foot and the pedal is constantly changing with each bump.

With **cleats** you can pull up on a pedal. Making it easier to put more energy into the cycle.

The cleat attached to the pedal in a solid position with a spring mechanism on the pedal that clamps.



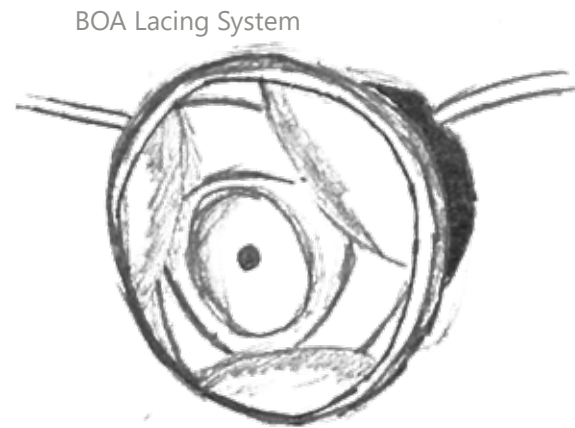
SPD-SL Compatible Pedal



There are many types of cleats. The cleat I will be selecting is **SPD-SL**, reasoning on **Page 24**.

The cleat should be positioned with the top at the ball of the foot for most comfort.

Strap must be able to tighten and loosen



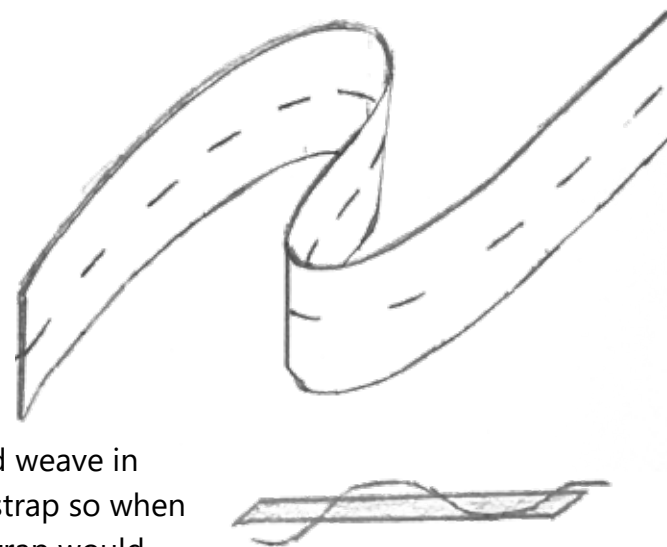
BOA Lacing System

Using a **BOA lacing system** [7] you can have a precise and reliable way to secure the strap.

It evenly tightens throughout the whole wire unlike regular laces, and will also not involuntarily come undone.

It is a **very fast** and **easy to use** system. To tighten, turn the dial, and to loosen the tension simply lift the dial.

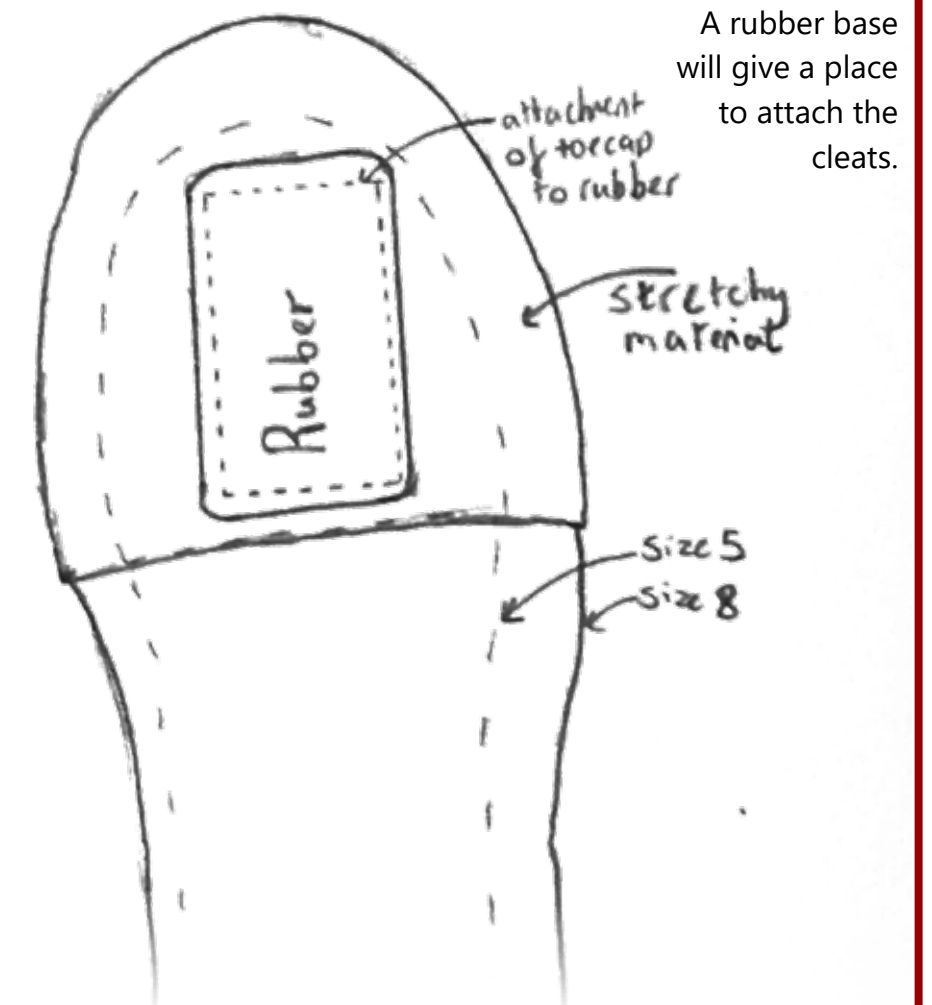
Durable and quick to adjust, it's ideal for cyclists who need stability and easy mid-ride adjustments.



The wires would weave in and out of the strap so when tightened the strap would scrunch up evenly.

Must be able to fit different shoe sizes

Using **neoprene's** stretchiness for the material around the toecap will allow fitting of different sizes.



A rubber base will give a place to attach the cleats.

Rubber plate would need to be smaller than minimum size requirement

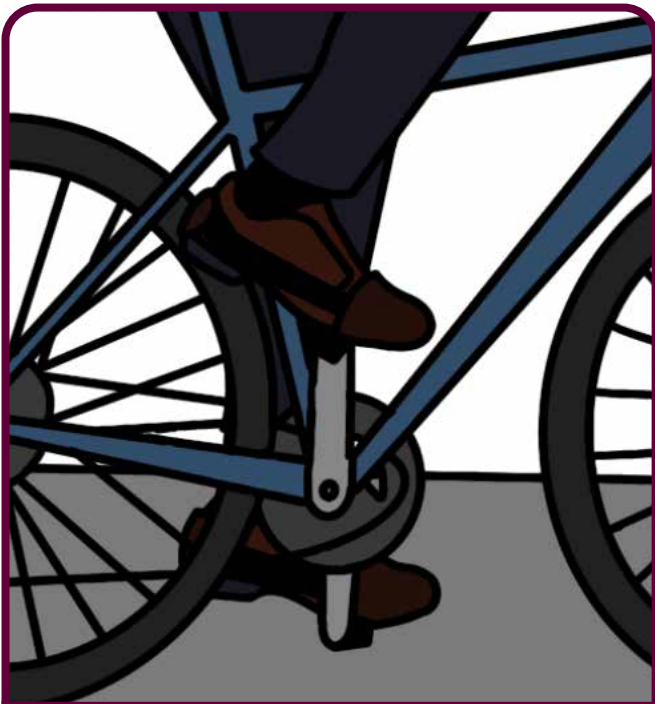
Neoprene is **durable, flexible and elastic**. Commonly used in cycling gear and wetsuits. Perfect for this application. [8]

Storyboard

Storyboarding of how the attachment would be used

Storyboarding Design and Development see Pages 25 & 26

1



Cycling on way to work

2



Getting to work and detaching

3



Working at their desk

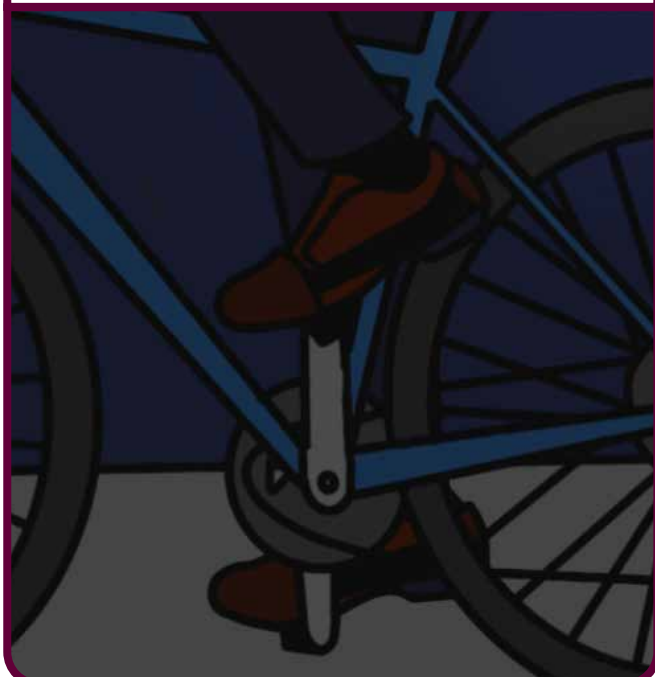
4

Leaving work and attaching



5

Cycling home



6

Getting home and taking rest



Benchmarking

Elastic Sole Protection [9]



Cleat Covers [10]



Criteria	My Concept	Road Cycling Shoe	Office Shoe	Elastic Sole Protection	Rubber Cleat Cover	Analysis
Detachable	Yes, the attachment can be removed	No	No	Yes, the attachment can be removed	Yes, the attachment can be removed	The detachability makes it practical, but I could explore making attachment and detachment even quicker and easier.
Can look like an office shoe	Yes, attaches onto office shoe	No	Yes	No, attaches onto cycling shoes	No, attaches onto cycling shoes	This is a key benefit, though I could work on minimizing the bulk further for a more seamless look.
Sole must connect to bike pedals securely.	Yes, using cycling cleats	Yes, using cycling cleats	No, flat soles with little grip	Yes	Yes	It offers good pedal connection similar to cycling shoes, but I need to ensure the cleats don't interfere with walking.
Must be able to tighten and loosen.	Yes, using BOA lacing	Yes, using BOA lacing	Yes, using laces	No	No	The BOA system works well, but I should test it thoroughly to make sure it holds up over time and feels comfortable.
Must be able to fit different shoe sizes.	Yes, fits varying shoe sizes	No, one product fits one size	No, one product fits one size	Yes, stretchy material can wrap around	Yes, attaches to cleat not shoe	The adjustable fit is a strong point. However, I need to ensure the material stays durable after frequent use.
Strap must be durable and securely attach.	Yes	N/A	N/A	No, can tear	N/A	The strap's durability should be tested under different tensions to confirm it doesn't loosen or wear out too quickly.
Must not damage the original shoe.	Yes, no abrasive parts touch the shoe.	N/A	N/A	Yes, no abrasive parts involved	Yes, attaches onto the hard cleat.	Protecting the original shoe is important. I should verify that the toecap and strap don't mark or degrade sensitive materials.
Toebox must be made of stretchy and durable material.	Yes	No, toebox does not change size	No, toebox does not change size	N/A	N/A	The neoprene toe cap is adaptable and strong, but it'll need testing in wet and varied conditions to ensure it performs consistently.
Must not exert too much pressure on wearer.	Yes, uses cushioning	Yes	Yes	Yes	Yes	The cushioning adds comfort.
Must not weigh too much.	Yes	No	No	Yes	Yes	Its lightweight design is an advantage. I'll make sure this doesn't compromise durability.
Must be able to be stored compactly.	Yes, all parts are compressible.	No, very rigid.	No, rigid	Yes, small volume and compressible.	Yes, small volume1	The compactness is a major feature, though exploring foldable or low-profile cleats could make storage easier.
Aesthetically pleasing	No, added bulk to a sleek shoe	No, not used for their aesthetics	Yes, professional, sleek and timeless	No, added bulk to a cycling shoe	No, added bulk to a cycling shoe	The bulkiness is a downside. I could make it integrate more seamlessly and look sleeker, or use different materials.

Morphological Analysis

Exploring different designs to fulfill requirements

Requirements	Idea I	Idea II	Idea III	Idea IV
Must be able to be stored compactly	Straps fold into the toecap and are secured with a snap button	Toecap folds flat	Toecap has a telescopic construction that can collapse	Everything can be dismantled with clips
Aesthetically pleasing	Matte black simple finish, similar to a road cycling shoe	Brown leather finish, similar to an oxford shoe	Streamlined and "aerodynamic" with no separation between parts	Customizable panels to fit user needs
Must not exert too much pressure on wearer	Thicker straps to spread pressure	Ergonomic heel cap that conforms to shoe	Gel padding inserts in heel strap	Thick memory foam heel cushioning
Must be able to tighten and loosen	Polished and elegant belt buckle	BOA lacing system (used on cycling shoes and outdoor gear)	Simple clip and tension lock	Elastic cord with tension lock

Selected Combination

Idea I:
Toecap Case

Idea III:
Streamlined Look

Idea I:
Thicker Straps

Idea II:
BOA Lacing

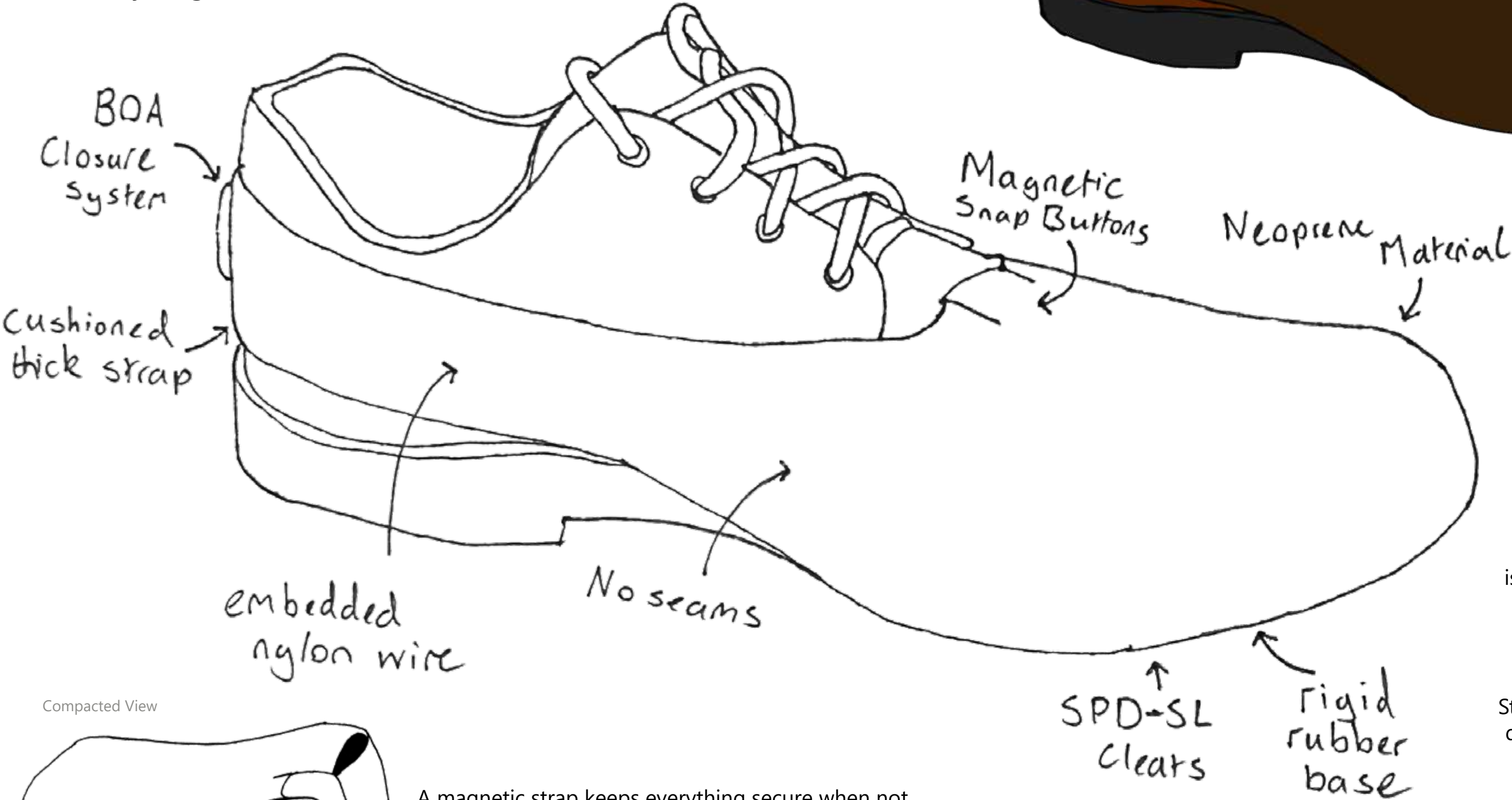
Final Design

For Final Design Development see Page 25 & 26

A removeable attachment for office shoes to enable wearer the use of cycling cleats.



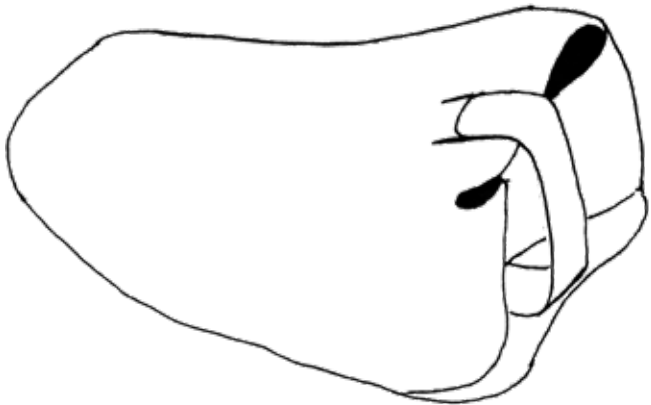
Coloured View



A stretchy neoprene toecap that stretches over the toes and extends along the sides to the heel. The fit is adjusted with a BOA closure system, using nylon wires integrated into the straps for tightening.

The bottom of the toecap is fitted with SPD-SL cycling cleats to road bike pedals. The design has no seams, for a smooth appearance and reducing weak points. Straps are padded to reduce discomfort when tightened.

Compacted View



A magnetic strap keeps everything secure when not in use. The toecap and its components can be folded neatly, with the strap holding them in place.

“Pedal-Ready”

Validation

Stretch

Using calculations to see if neoprene would be a suitable material for the toecap. As the main concern is the width stretching from a **UK size 5-12**, as the length is controlled by the BOA system to tighten, so will not need to stretch as much, only compress by folding in on itself.

Minimum Strain Required

Size 5 = 9.7cm

Size 12 = 12cm^[26]

$$\frac{12\text{cm}}{9.7\text{cm}} = 1.237 \dots \approx 1.24$$

$$\epsilon = \frac{\Delta L}{L_0} = 0.24$$

Stress Required

$$\sigma = E \times \epsilon$$

$$\sigma = 6.136 \times 10^6 \times 0.24$$

$$\sigma = 1.47264 \times 10^6$$

$$\sigma \approx 1.47\text{MPa}$$

Tensile Strength
Range of Neoprene

500 – 3000PSI ^[27]

$$500\text{PSI} \times 6894.76 \approx 3.45\text{MPa}$$

$$3000\text{PSI} \times 6894.76 \approx 20.7\text{MPa}$$

As the minimum stress is less than the range of neoprene (1.47MPa > 3.45MPa). Neoprene can safely stretch from a size 5 to 12. Therefore suitable for this purpose.

Mass

BOA System: The weight of the BOA dial and nylon wires typically falls between **10 and 20 grams** ^[20].

SPD-SL Cleats: A pair of cleats has a mass of approximately 72 grams ^[28], meaning **36 grams** each.

Neoprene: Using an estimated thickness of **2mm** for the neoprene material and applying its density (**1.23g/cm³**^[8]), the mass of the upper part of the shoe is calculated to be around **183 grams**. This estimate will slightly overstate the mass since it uses a cuboid approximation for calculating the area (shown on **Page 29**).



Adding these components together gives a total estimated mass of **229 to 239 grams** per shoe, which fits comfortably within the target mass of under **250 grams**.

Summary and Improvement

This confirms that it is technically feasible based on calculations, and is desirable according to feedback. Though it is worth noting that several assumptions were made, such as the thickness of neoprene and the surface area. Additionally, the user feedback was not exclusively from the target audience, and the sample size of 42 participants is relatively small. Furthermore, the lack of a prototype for the final design means potential flaws or considerations might have been overlooked. Despite these limitations, the overall results indicate that the concept is both technically viable and well-received by users.

Feedback

I collected third-party feedback using a Microsoft Form, receiving a total of **42** responses. The questions in the form focused on evaluating two key aspects: desirability and ease of use.

1. How desirable do you think this product would be for a commuter?

4.31

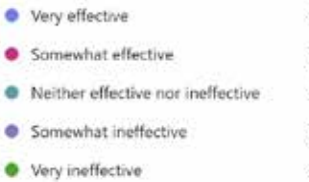
Average Rating



2. How effective do you think the BOA system is?

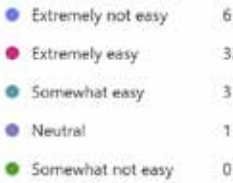
Info: BOA Tightening System
(commonly used on outdoor gear and bike shoes)
<https://www.boafit.com/en-gb>

[More details](#)



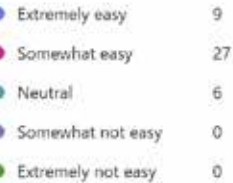
3. How easy do you think this product is to put on?

[More details](#)

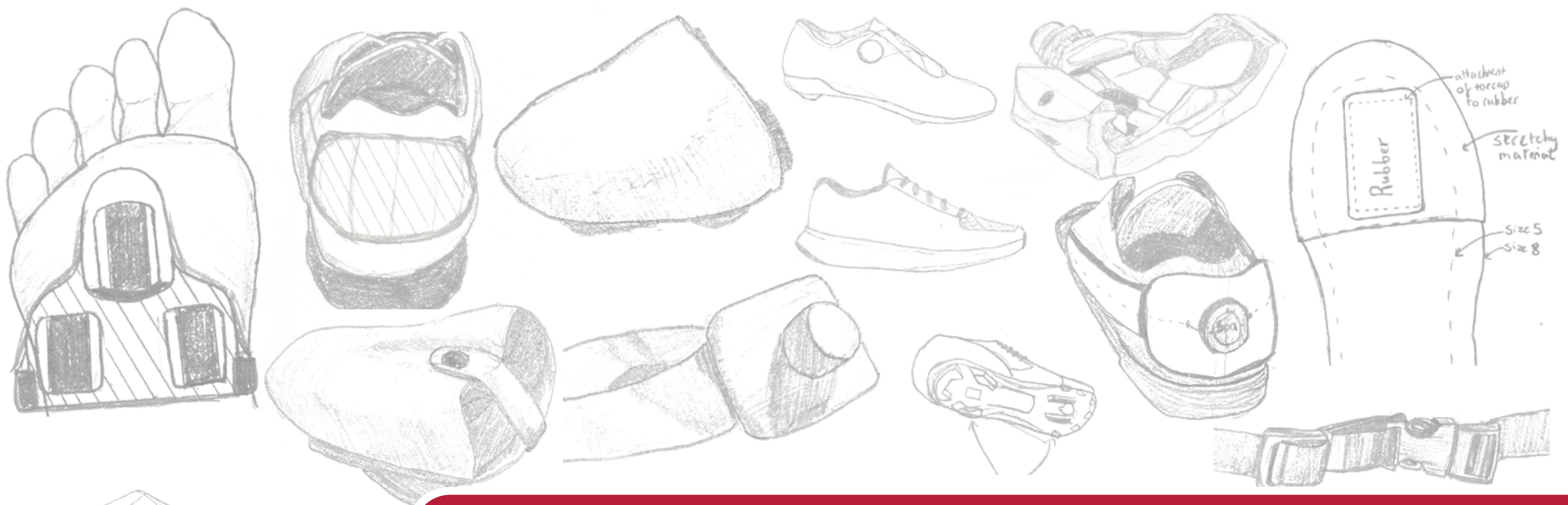


4. How easy do you think the product is to collapse and become compact?

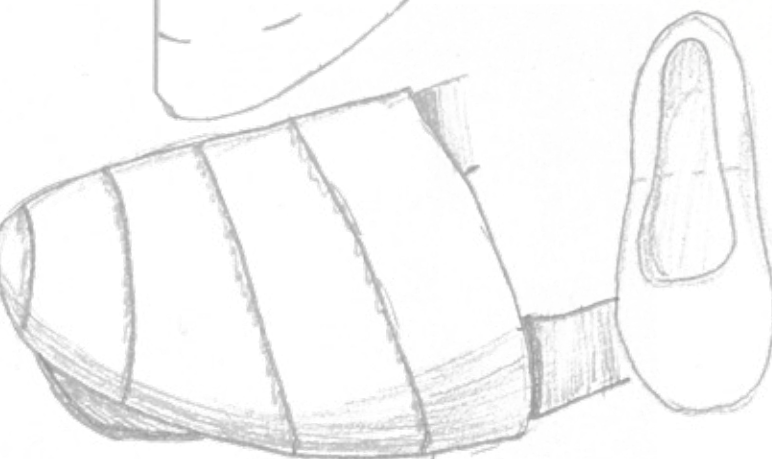
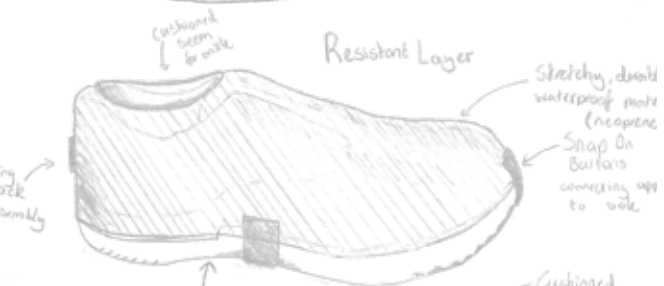
[More details](#)



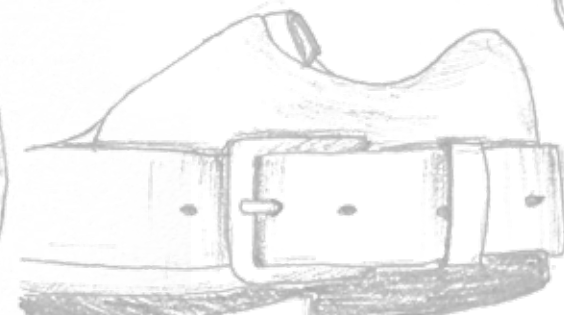
The feedback was largely positive, with respondents giving an average rating of **4.31 stars** for desirability. Similarly, the questions related to ease of use received a majority of **4-star** ratings, indicating that most participants found the concept intuitive and user-friendly. Though there are drawbacks with this, all of the feedback came from students, and not all regularly cycled.



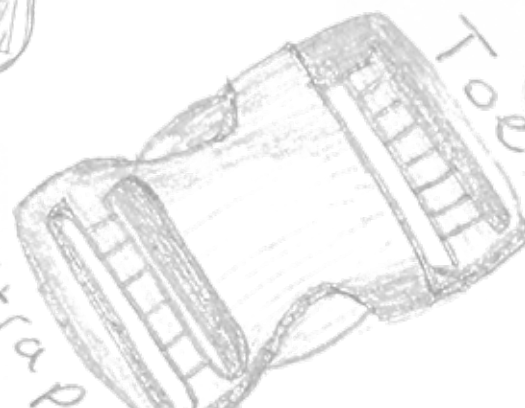
Supplementary Pages



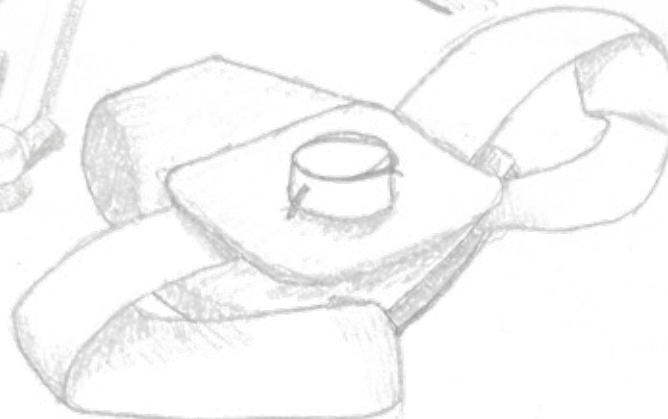
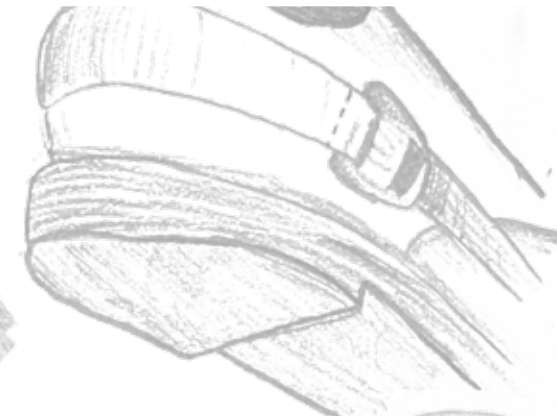
Flexible/Foldable Design



Expandable cushioning/grip

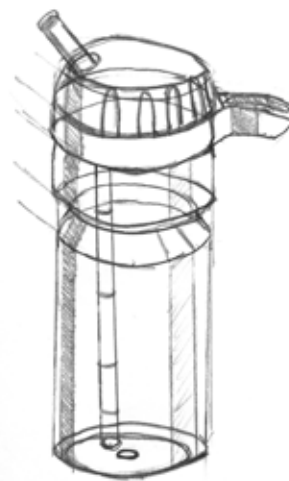
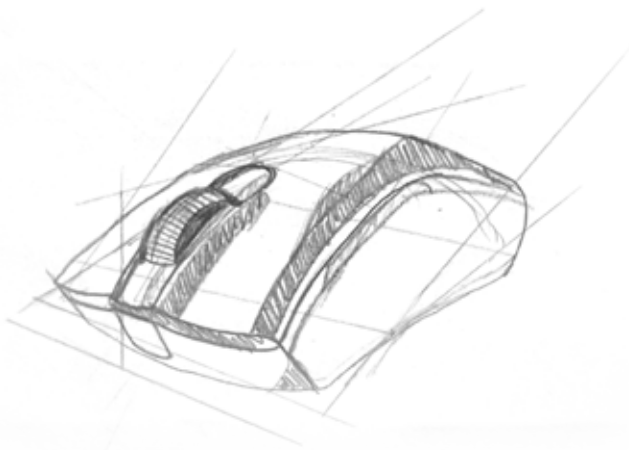
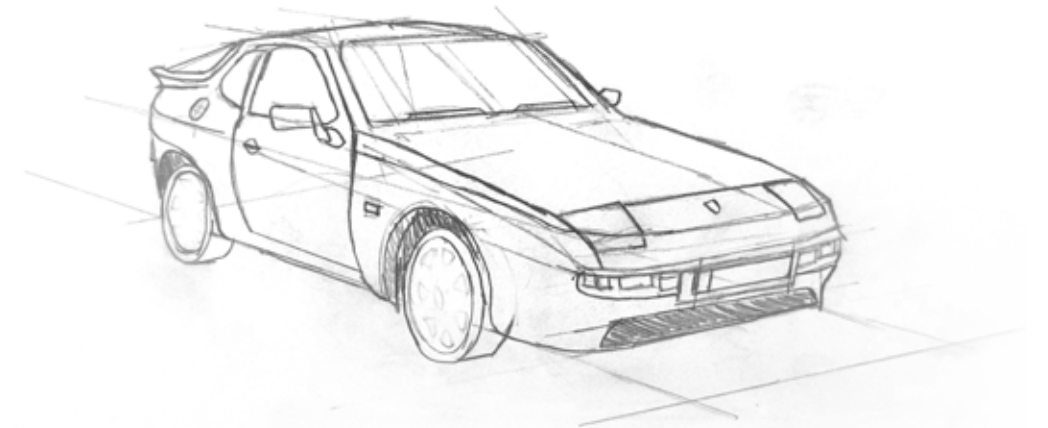
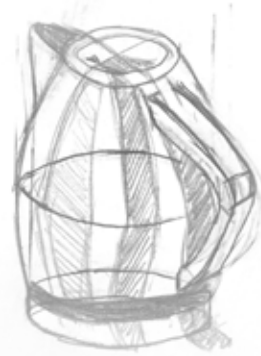
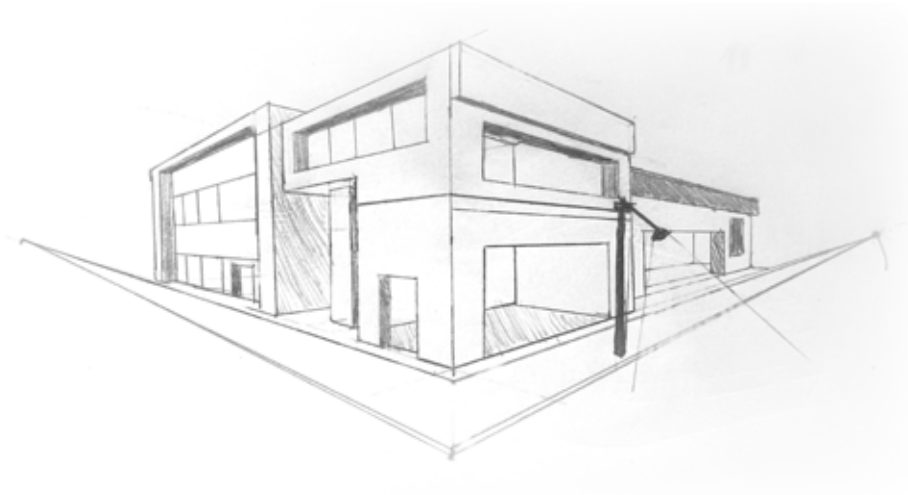


Toe



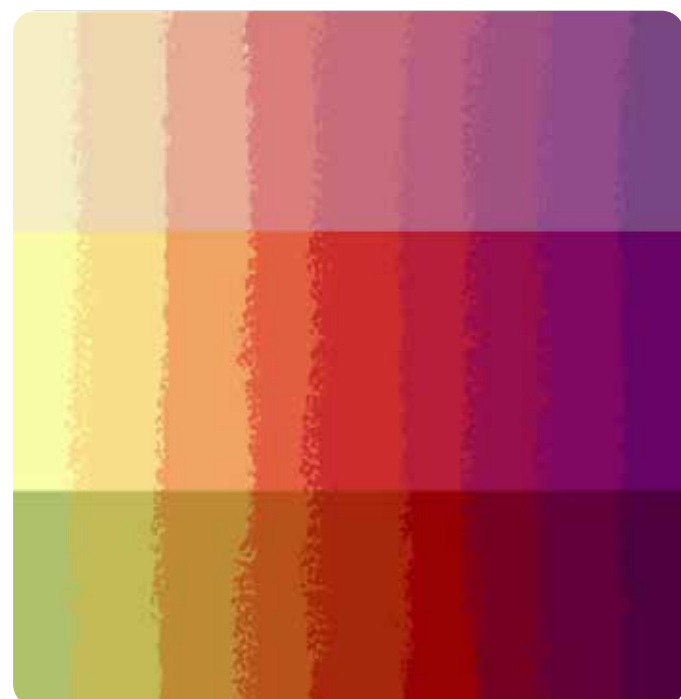
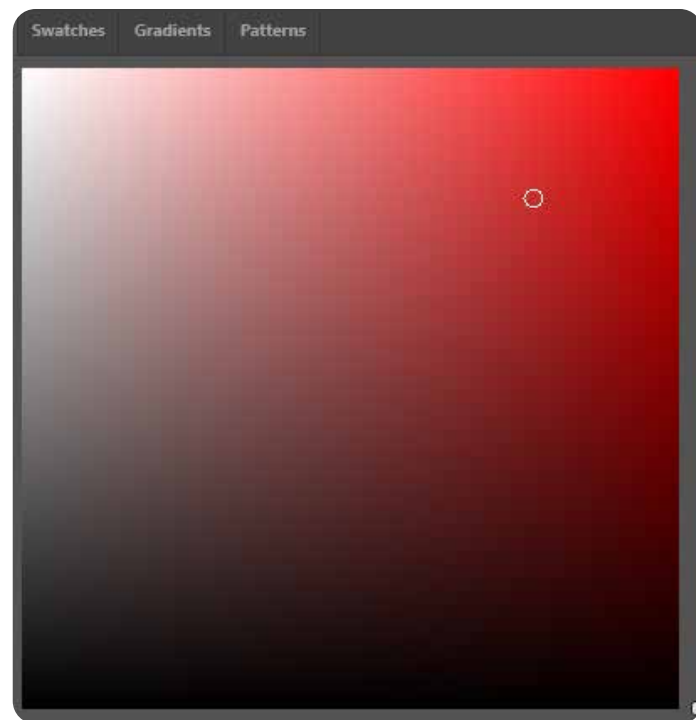
Sketching Practice

For easier viewing, all sketches were imported into photoshop where they were set to black and white and the colour curves were adjusted so the background was white. Later sketches were adjusted further to include a transparent background

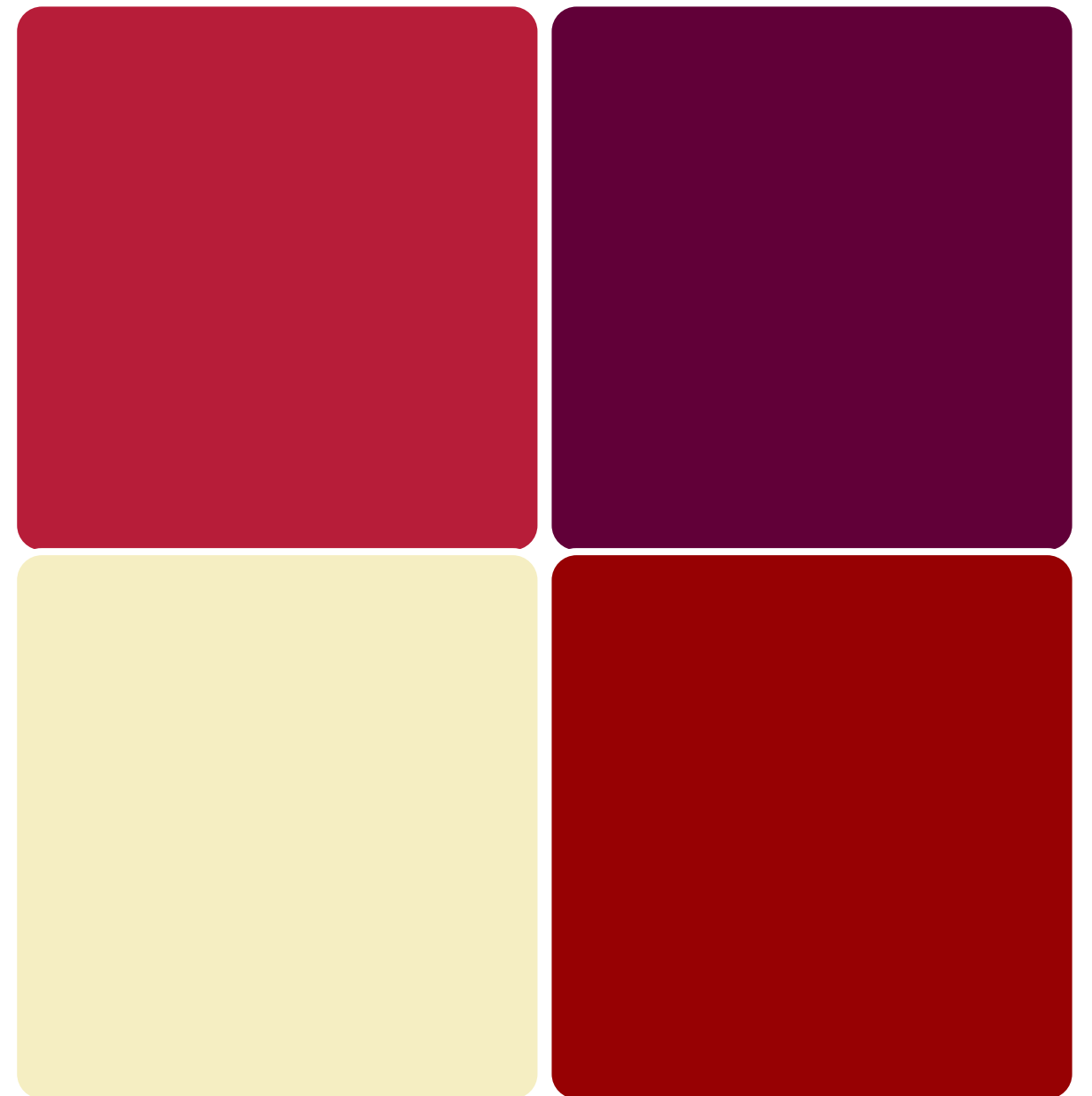


Colour Pallet

In Photoshop, I started by selecting a base red color and created a gradient by adjusting the brightness and saturation along a curve. The curve transitioned from the lightest, least saturated red to the darkest, most saturated shade.



From this gradient, I chose a new color and adjusted the hue to build a wider gradient range. To give the gradient more depth, I divided it into three distinct sections: one third brighter and less saturated for pastel tones, the middle section as a normal intensity, and the final third darker and more saturated for deeper shades.

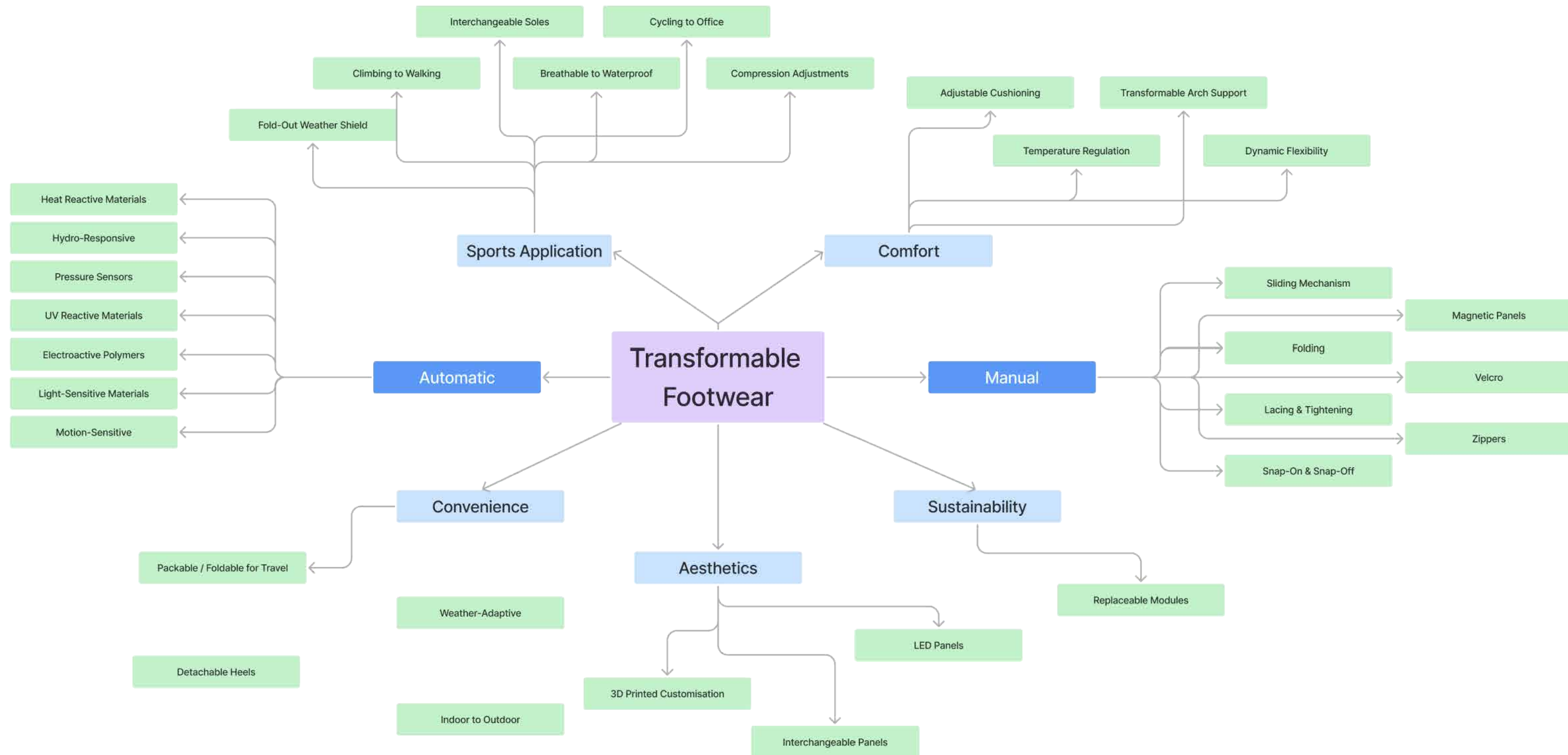


Once the gradient was complete, I selected 3 to 5 colors from across these sections to construct cohesive palettes with variations in tone and saturation.



Mindmap

Personal mindmap used for ideation



Lo-Fi Prototyping Practice

For my lo-fi prototyping exercise, I chose to create a “smart shoe” aimed at office workers, based on an oxford shoe. I completed the prototyping in roughly two hours, using paper, cardboard, and foam to construct a mockup.



For the sole. I cut out two identical sole templates from thick cardboard, which I attached to both sides of a foam block. Using a hot wire cutter, I trimmed the foam around the templates to match the desired outline, then filed the foam to create a slight heel and arch, shaping the shoe’s overall profile.



Next, I assembled the upper part of the shoe. I cut out three main pieces from paper: a large “U” shape for the front, covering the toe area, and two side pieces designed to resemble the panels found on a classic oxford shoe. I made sure each piece had additional material at the bottom edge to allow for secure attachment to the base. Using double-sided tape, I fixed each piece to the cardboard base, folding and curving them to give the shoe a more natural shape. I then used tape to join the paper components, creating the semblance of a continuous upper, and cut small holes in the side pieces for shoelaces.

I then attached the upper shoe and the foam sole together using hot glue for a secure bond.

Reflection:

The toe area turned out to be steeper than intended and the back could benefit from being smaller and more curved. As the current shape gives the shoe a triangular appearance, which I did not intend.

BOA System

The BOA lacing system is an advanced closure mechanism designed to provide precise, adjustable, and secure fitting. It has three main components: a micro-adjustable dial, nylon laces, and low-friction guides.

To use the system, push the dial in, and turn it to tighten or loosen the fit, and pulling it up to quickly release the laces. This mechanism allows for incremental adjustments, ensuring a comfortable and supportive fit that can be modified on the go without the need for manual knotting or tying laces. [7] [25]

Common Uses:

- Sports shoes
- Cycling shoes
- Snowboard boots
- Ski boots
- Braces
- Prosthetics
- Orthopedic shoes
- Work boots
- Protective helmets
- Fishing waders
- Climbing gear



Dial:

The central component is a micro-adjustable dial. By turning the dial clockwise, it tightens a specialized lace or cable. Turning it counterclockwise loosens the lace, allowing precise adjustments to the fit.

Laces/Cables:

Instead of traditional laces, the BOA system uses lightweight, durable, and low-friction cables, often made of materials like stainless steel or nylon.

Guide Channels:

The cables are routed through a series of low-friction guides or eyelets on the shoe or gear. These channels ensure even tightening across the entire closure area.

Lock-and-Release Mechanism:

The dial locks in place to maintain tension during use. To release the system, you typically pull the dial outward, allowing the cables to loosen quickly. [7] [25]



"BOA® laces are designed to withstand loads up to ten times higher than those required for a tight fit." [7]

Prototyping Development

Both the collapsible shoe and the resistant layer cover went through two iterations: one created before obtaining a real shoe and another after.

The collapsible shoe was created using a thin foam sheet cut into the outline of a shoe with a slightly pointed front. Four layers of paper were taped on top, with the edges sanded down to align seamlessly with the foam without any sharp cuts. A hinge was created by slicing through the foam in the middle, leaving the paper layer intact. However, I chose to redo this prototype as the overall quality was unsatisfactory, and the hinge was weak, quickly wearing out with use.

The resistant layer cover was initially constructed using cardboard, which was cut and folded to mimic the shape of a shoe. The sole was created from thick foam, cut to match the shoe's outline, and then sliced horizontally to form a heel. These two foam pieces were then layered together. I revisited this prototype because the cardboard lacked the desired texture and flexibility.

Initial Prototype



Cut Soles



Initial Prototype



Hot Wire Cutting



Feux Leather used in actual prototype



Paper used in initial and actual prototyping



Double Sided Tape



Cycling Cleat Choices

Mountain Bike Cleats

Designed for rugged, off-road conditions such as mountain biking or cyclocross. Features like mud-shedding and recessed cleats are unnecessary for urban commuting. They are heavier and less efficient for power transfer, which is critical for road cycling.

SPD Cleats [16][17][18]

Small, recessed cleats that are durable and easy to walk in.

Strengths:

- Double-sided engagement for easier clipping in and out.
- Excellent for off-road use due to mud-shedding capability.
- Compatible with many cycling shoes.

Weaknesses:

- Smaller surface area reduces power transfer efficiency.
- Heavier and bulkier, making them less practical for road cycling and commuting.



Speedplay Cleats [16][17]

Unique circular cleats with dual-sided engagement.

Strengths:

- Adjustable float and tension for ergonomic benefits.
- Easy to clip in from either side of the pedal.

Weaknesses:

- More expensive than other cleats.
- Requires frequent maintenance.
- Less accessible and practical for casual or commuter riders.

Look Keo Cleats [16][17]

Lightweight, triangular cleats for road cycling.

Strengths:

- Similar power transfer and efficiency as SPD-SL cleats.
- Adjustable float options for rider comfort.

Weaknesses:

- Compatibility limited to Look pedals.
- Faster wear due to plastic construction.
- Higher cost compared to SPD-SL cleats.



SPD-SL Cleats [16][17][18]

Larger, plastic cleats with a three-bolt system.

Strengths:

- Superior power transfer due to a larger contact area.
- Lightweight design optimized for road cycling.
- Widely compatible with road bike pedals.

Weaknesses:

- Protruding design makes walking difficult.
- Plastic material wears down faster with use.
- Single-sided engagement requires precise clipping.

Why Use SPD-SL Cleats?

Ideal for Commuting:

Optimized for road cycling, offering excellent power transfer and stability.

Lightweight and compatible with many road bikes, making them accessible to urban riders.

One of the most common cycling cleats.

Product Integration:

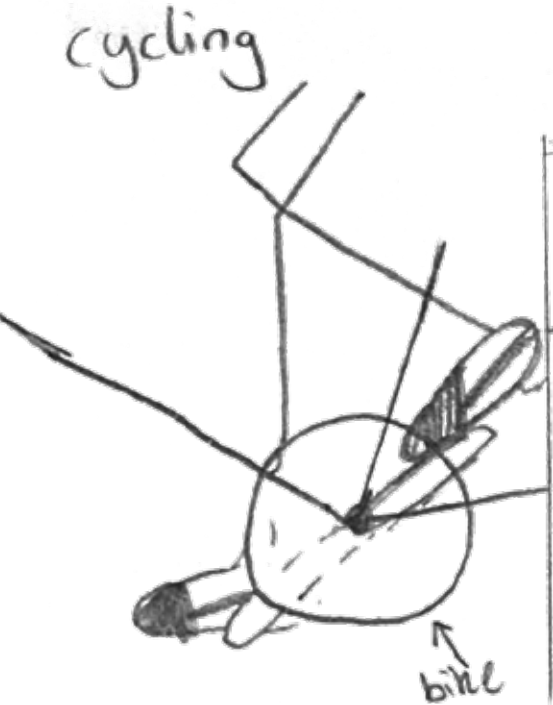
While SPD-SL cleats are less suitable for walking the user can simply remove the attachment to walk.



Initial Storyboarding

Simple initial storyboarding design with stickmen

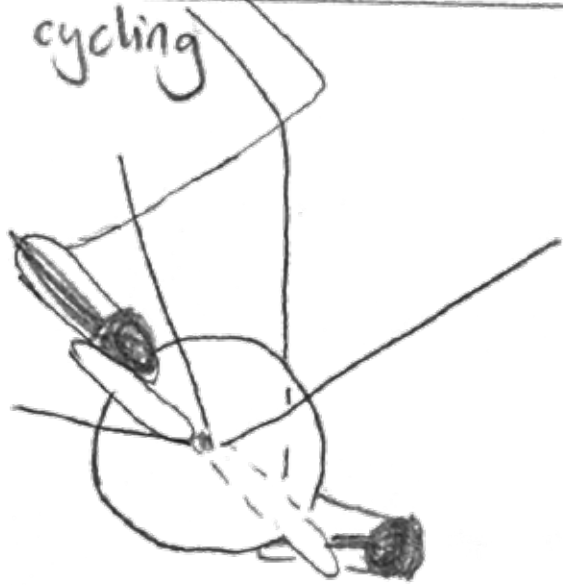
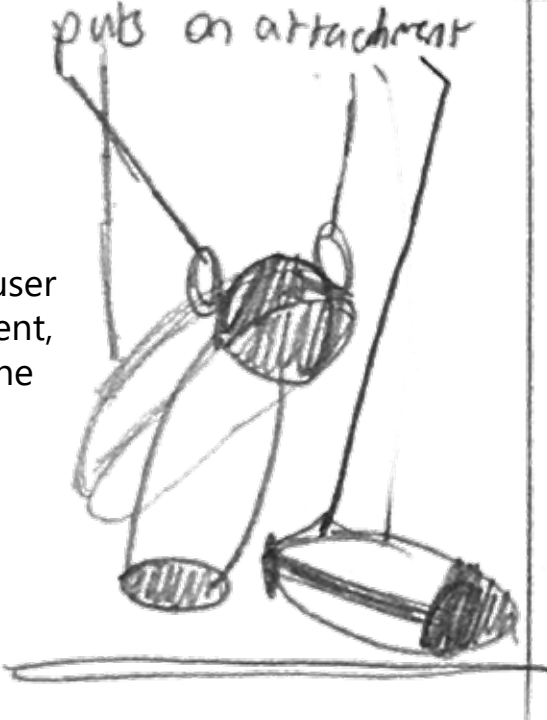
Focus is on the shoes in all frames



Scene 2 shows how a user would detach, with the user adjusting the BOA



Scene 5 shows how a user would put on attachment, with the toe first and the strap hanging down

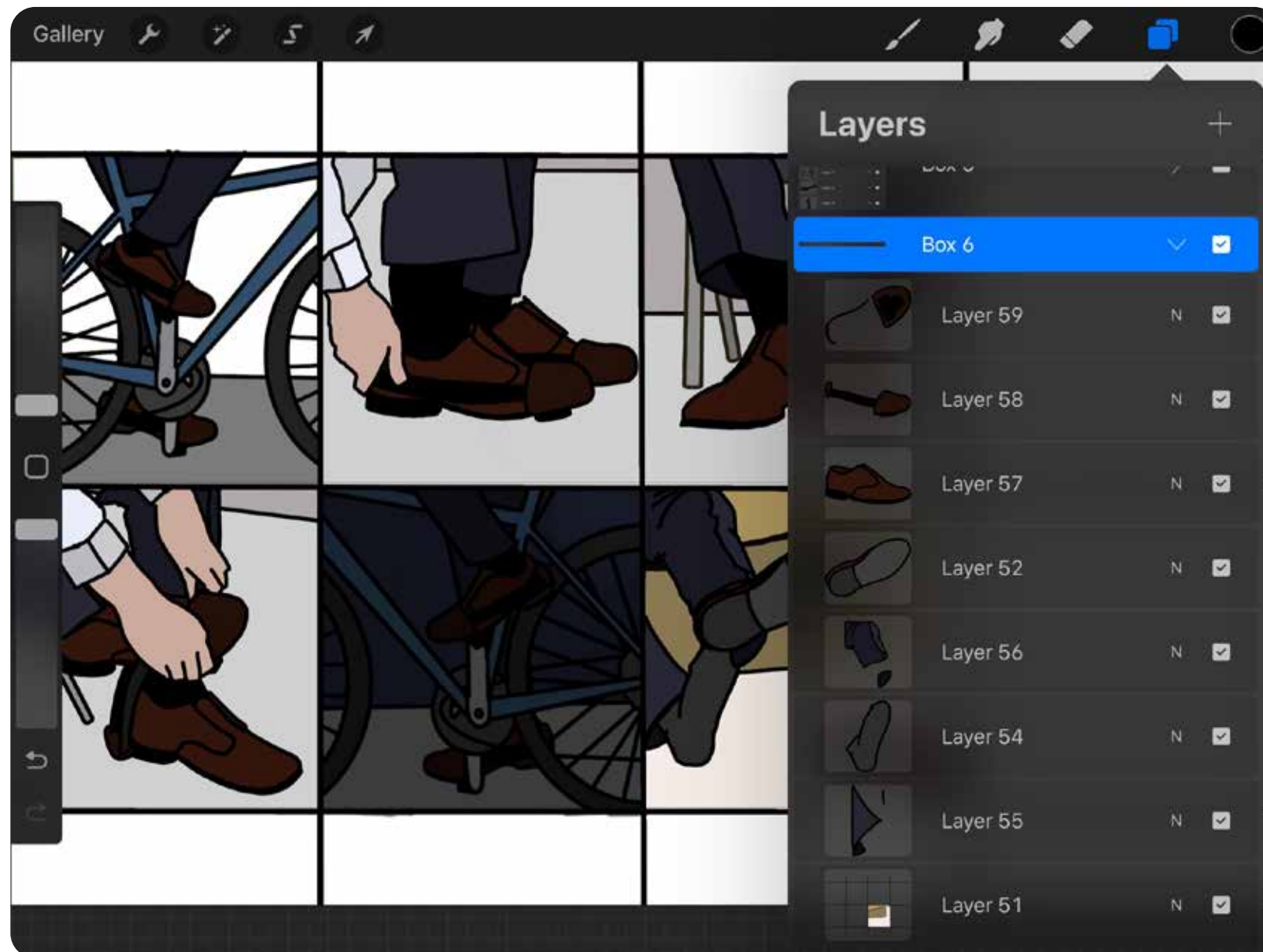


Cycling scene is flipped to show a different direction of travel (from work to home)



Using Procreate

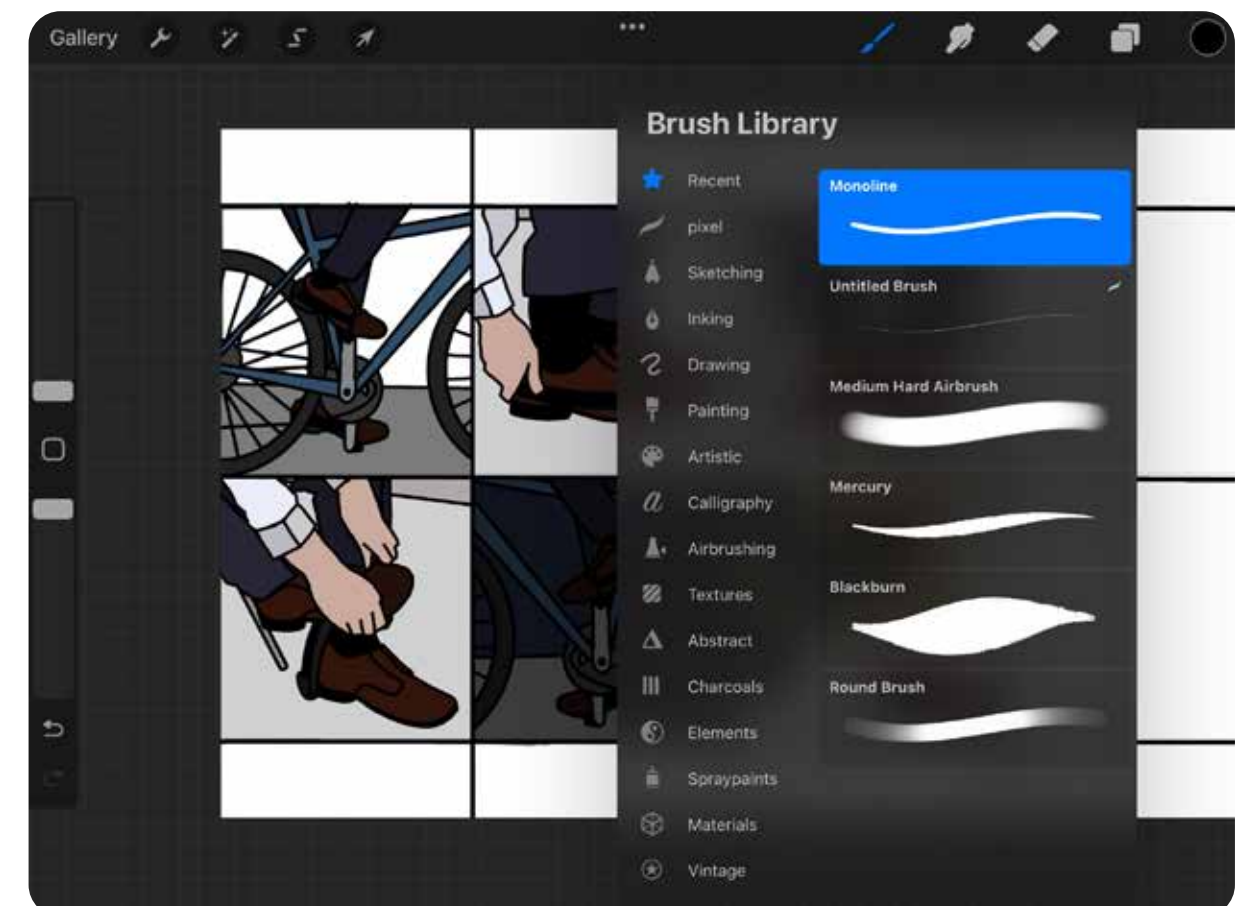
Developing my storyboard using procreate



Using Procreate, I created a grid and designated each box as its own section to work on. I utilized layers to organize the components effectively. I started by sketching each element in full detail and then moved on to adding color. To ensure a realistic appearance, I carefully arranged the layers in a logical order. However, I decided not to include shading since even applying flat colors took several hours. Additionally, I felt that shading wouldn't add significant value to the overall piece.

The project ended up consisting of 59 different layers across six boxes. For some scenes, I took reference photos of myself, especially for poses that were complex or unique. This was particularly helpful for capturing intricate details, such as hand gestures and specific shoe placements.

I had to make custom brushes for some situations, such as a singular pixel brush for filling in colour in tight spaces, or for cleaning up.



Final Design Development

Design improvements from morphological analysis to final design

In my morphological analysis, I explored various ideas to meet the key design requirements and ultimately decided on using a toecap case, a streamlined appearance, thicker straps, and a BOA closure system.

Building on these choices, I refined the design further. The streamlined appearance was simplified to create a more elegant look with all one material and colour. The inclusion of thicker straps complemented this aesthetic. Initially, the toecap case used snap-on buttons, but I felt this approach was too basic for the design's intended appearance. To enhance the look and functionality, I opted for magnets.

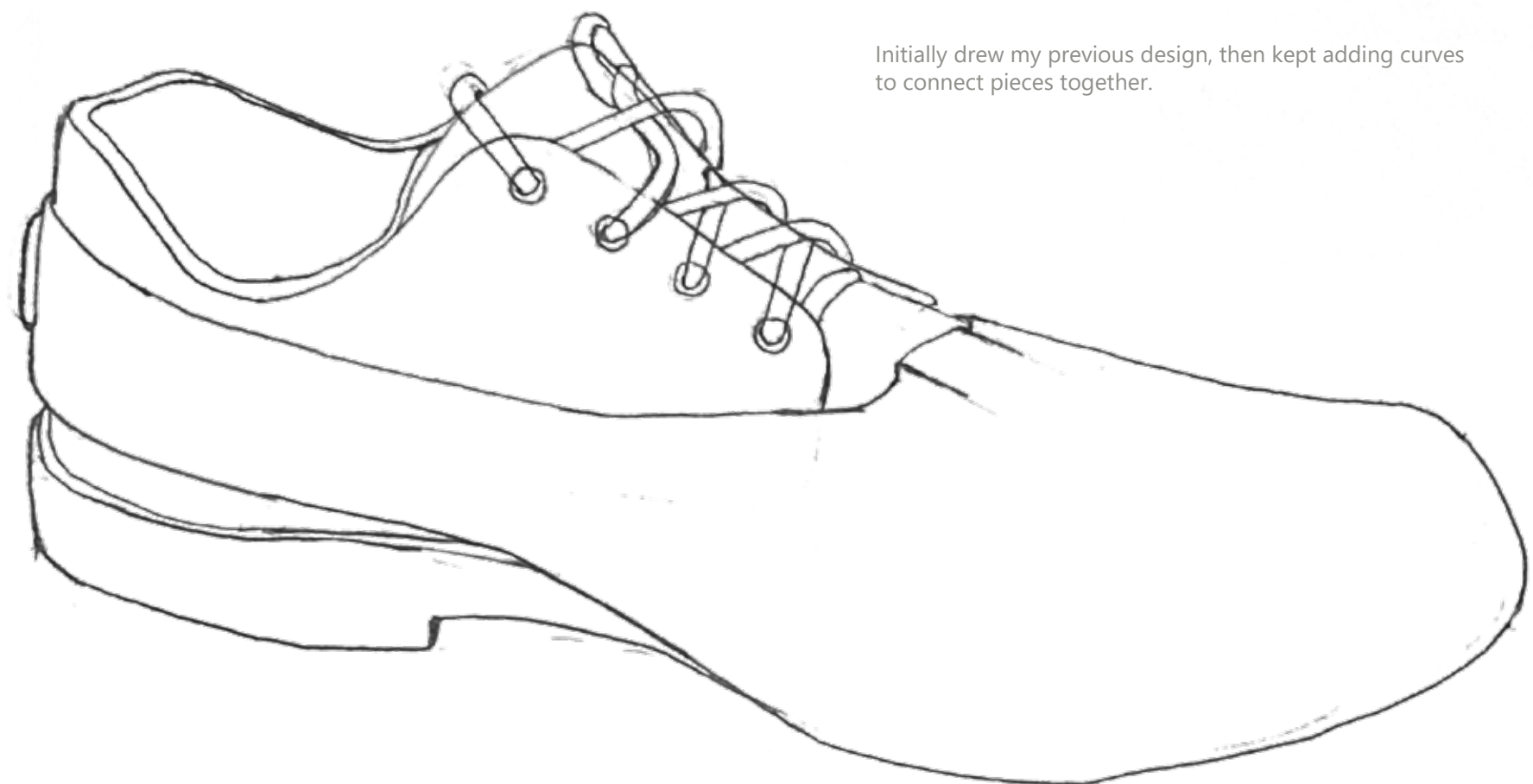
For the magnetic strap; while worn, the strap is concealed inside the sole of the toecap. For storage, it can be pulled out and secured to the top of the toecap. For the sole, I decided that instead of placing the rubber base externally, it would be more effective and visually appealing to embed it inside the bottom of the toecap. Similarly, the nylon wires of the BOA system were integrated into the straps to achieve a cleaner design.

Finally, I opted for the straps to not only be thicker but also padded, combining ideas I, III, and IV from my morphological analysis. This combination improved both the comfort and functionality of the design.

Started off with a simple oxford shoe



Initially drew my previous design, then kept adding curves to connect pieces together.

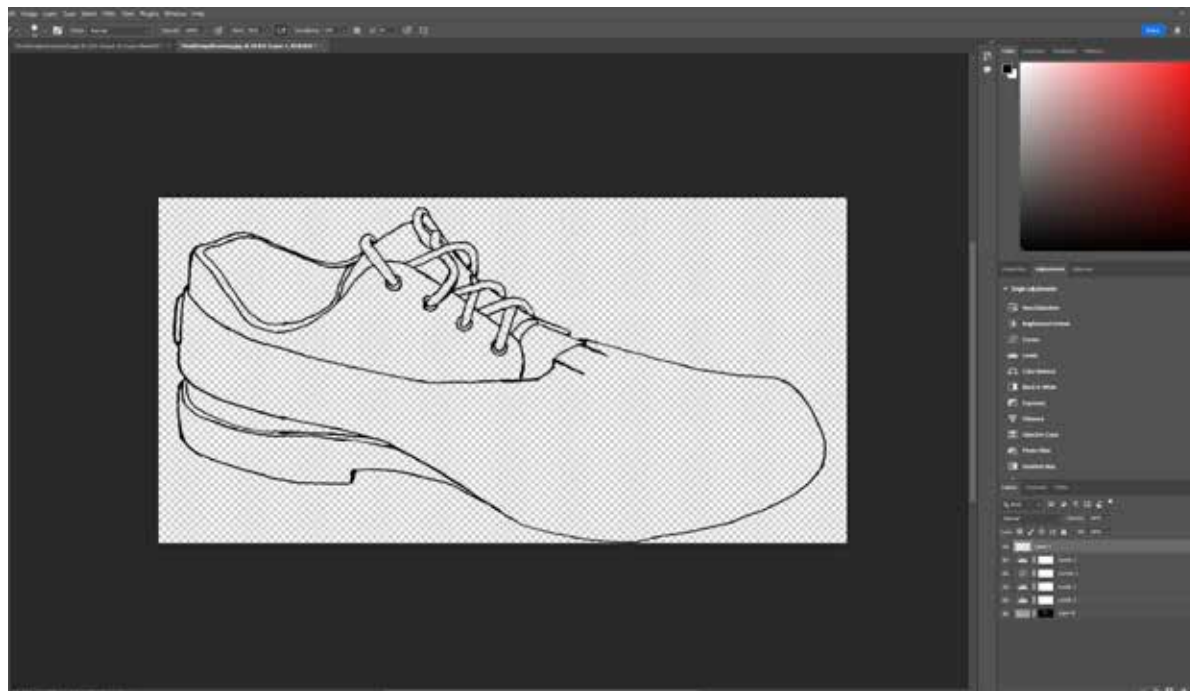


Learning Illustrator

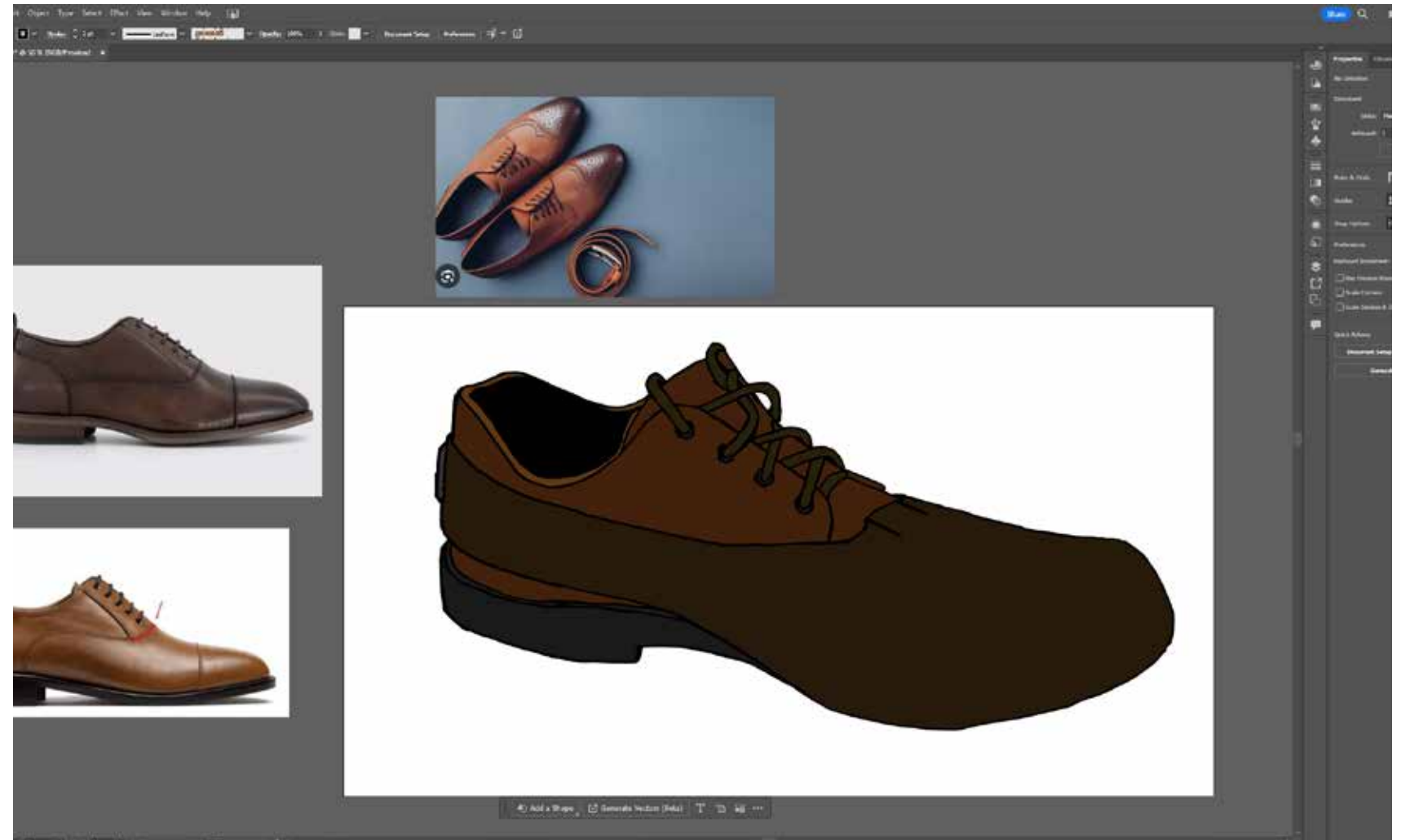
Using illustrator and photoshop for sketches

As I drew all of my designs on a paper sketchbook with pencil, it took a lot of refining to get it to look good digitally.

Then using the function image trace, I could convert all of my pencil marks into strokes. I could then use the live paint tool to colour drop into the white spaces.



I took scans of my sketches then used photoshop to edit the curves. Essentially turning anything not black into white. Then using colour range I could mask out any white so I was just left with the black sketches



I used many image references to choose the colourings for my design. I thought that the brown shoe was the most versatile and that black was too dull.

Validation Calculations

Calculations were then re-written using an online text editor for equations, iMathEQ [29]

Stretch Calculation for Neoprene

UK size 5-12

UK size 5: 9.7cm

UK size 12: 12cm

Stretch ratio

$$\frac{12}{9.7} = 1.237...$$

1.24 (to 3 sig figs)

Neoprene would need to stretch to 1.24 times its width.

$$\epsilon = \frac{\Delta L}{L_0} = 0.24$$

$$\sigma = E \times \epsilon$$

$$\sigma = 6.136 \times 10^6 \times 0.24 =$$

$$\sigma = 1.47264 \times 10^6 \text{ Pa}$$

$$1.473 \text{ MPa}$$

Tensile stress range:

500-3000 PSI

Converting

$$500 \text{ PSI} \times 6894.76 \approx 3.45 \text{ MPa}$$

$$3000 \text{ PSI} \times 6894.76 \approx 20.68 \text{ MPa}$$

Weight Calculation for Neoprene

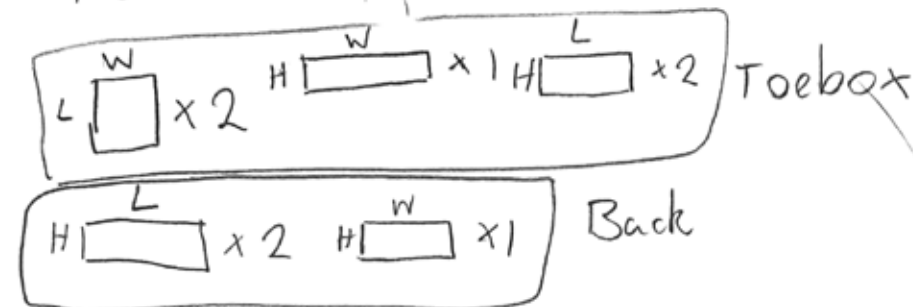
7cm height

Toebox width = 10cm

Toebox length = 12cm

Back length = 15cm

Back Width = 8cm



$$\begin{aligned} \text{Toebox} &= (10 \times 12) \times 2 \\ &+ 7 \times 10 \\ &+ (7 \times 12) \times 2 \\ &= 478 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Back} &= (7 \times 15) \times 2 \\ &+ 7 \times 8 \\ &= 266 \text{ cm}^2 \end{aligned}$$

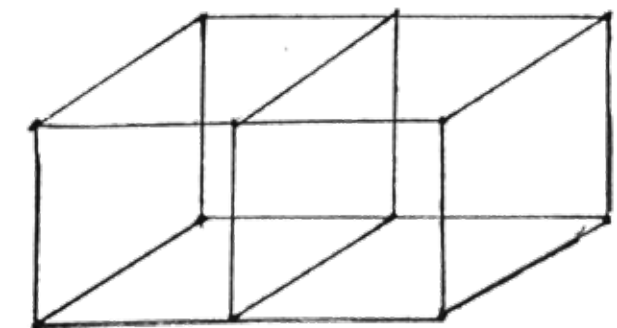
$$\text{Area} = 744 \text{ cm}^2$$

$$\text{Thickness} = 2 \text{ mm}$$

$$\text{Volume} = 148.8 \text{ cm}^3$$

$$\text{Neoprene density} \approx 1.23 \text{ g/cm}^3$$

$$\text{Weight} = 183 \text{ g}$$



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